

Horton Analysis Replica

```
##### ABOUT THIS SCRIPT #####
### This script reads in the clean dataset from data folder and then
### replicates the methods in Horton et. al to determine change.
```

```
#####|
##### LOAD IN NECESSARY PACKAGES #####
library(dplyr)
```

```
##
## Attaching package: 'dplyr'
```

```
## The following objects are masked from 'package:stats':
##
##   filter, lag
```

```
## The following objects are masked from 'package:base':
##
##   intersect, setdiff, setequal, union
```

```
library(here)
```

```
## here() starts at /Users/nis100/tbPrevalenceReviewLMIC/tb-lmic-prevalence-review
```

```
library(magrittr)
library(DT)
library(metafor)
```

```
## Loading required package: Matrix
```

```
## Loading required package: metadat
```

```
## Loading required package: numDeriv
```

```
##
## Loading the 'metafor' package (version 4.8-0). For an
## introduction to the package please type: help(metafor)
```

```
library(meta)
```

```
## Loading 'meta' package (version 8.0-2).
## Type 'help(meta)' for a brief overview.
```

```
library(brms)
```

```
## Loading required package: Rcpp
```

```
## Loading 'brms' package (version 2.22.0). Useful instructions  
## can be found by typing help('brms'). A more detailed introduction  
## to the package is available through vignette('brms_overview').
```

```
##
```

```
## Attaching package: 'brms'
```

```
## The following object is masked from 'package:stats':
```

```
##
```

```
##      ar
```

```
library(posterior)
```

```
## This is posterior version 1.6.0
```

```
##
```

```
## Attaching package: 'posterior'
```

```
## The following objects are masked from 'package:stats':
```

```
##
```

```
##      mad, sd, var
```

```
## The following objects are masked from 'package:base':
```

```
##
```

```
##      %in%, match
```

```
library(tidybayes)
```

```
##
```

```
## Attaching package: 'tidybayes'
```

```
## The following objects are masked from 'package:brms':
```

```
##
```

```
##      dstudent_t, pstudent_t, qstudent_t, rstudent_t
```

```
#####  
##### LOAD CLEAN DATA SET AND FILTER TO SURVEYS REPORTING SEX #####  
cleanSexDf0 <- readRDS(here("data/fullDataClean.rds")) %>%  
  filter(report.sex == "Yes")  
#####  
##### ADD WHO REGION AND STUDY ID TO DATA #####  
regionWHO <- read.csv("data/who-regions.csv")[,c(1,4)] %>%  
  rename(study.country = Entity,  
         WHO.region = World.regions.according.to.WHO)
```

```

figID <- read.csv(here("data/titlesForFigures.csv"))
colnames(figID)[1] <- "covidence.id"

cleanSexDF0 %<>% left_join(figID[,c("covidence.id", "figure.id")],
                          by = "covidence.id") %>%
  mutate(figure.id.yr = paste(figure.id, study.years))

### Rename some countries to match the WHO CSV.
### For labeling these will remain as extracted.

cleanSexDF0[which(cleanSexDF0$study.country == "Viet Nam"), "study.country"] <- "Vietnam"
cleanSexDF0[which(cleanSexDF0$study.country == "The Gambia"), "study.country"] <- "Gambia"
cleanSexDF0[which(cleanSexDF0$study.country == "United Republic of Tanzania"), "study.country"] <- "Tanzania"
cleanSexDF0[which(cleanSexDF0$study.country == "Lao PDR"), "study.country"] <- "Laos"
cleanSexDF0[which(cleanSexDF0$study.country == "Democratic People's Republic of Korea"), "study.country"] <- "North Korea"

### Join the WHO data
cleanSexDF0 %<>% left_join(regionWHO)

```

```
## Joining with 'by = join_by(study.country)'
```

```

### Open a PDF
# pdf(file = here("output/HortonAnalysis.pdf"), width = 8.5, height = 11.5)

### Calculation of the bacteriological TB prevalence among males
### Random effects weighted model
### Option 1: Use only those surveys which report counts of
### participants and bacteriological TB.
prevSexBact <- cleanSexDF0 %>%
  filter(is.na(n.bacteriological.tb.male)== FALSE &
         is.na(n.participants.male)== FALSE)

### Males
propBactMale <-metaprop(n.bacteriological.tb.male, n.participants.male, figure.id.yr,
                       data=prevSexBact, random=TRUE,
                       pscale = 1e5)

summary(propBactMale)

```

##	events
## India (Gujarat State) 2011-2012	550.2381
## India (Wardha District) 2007-2009	242.7571
## India (Saharia Tribal) 2015-2015	3170.2128
## South Africa (Western Cape Province, a) 2005	2567.5676
## Viet Nam (National) 2006-2007	497.6175
## Malawi (National) 2013-2014	481.3202
## China (Yunnan Province) NA	1625.8400
## Kenya (Asembo Area) 2006-2007	763.6603
## Phillipines (National) 2007	963.7035
## India (Faridabad District) 2008-2009	138.2771
## Uganda (Rubaga) 2008-2009	439.5036

## India (Karachal) 2007-2008	2123.6610
## India (Jabalpur District) 2009-2010	353.3936
## Pakistan (National) 2010-2011	403.8916
## Kenya (Karemo) NA	310.1978
## The Gambia (National) 2012-2013	279.9360
## Tanzania (National) 2011-2012	434.0487
## Nigeria (National) 2012	473.0994
## Myanmar (National) 2009-2010	919.8893
## Cambodia (National) 2002	1915.5144
## Lao Pdr (National) 2010-2011	885.1054
## Viet Nam (Hanoi City) 2003-2004	230.4147
## Rwanda (National) 2012	164.8805
## India (Chennai) 2010-2012	501.0578
## India (Madhya Pradesh Tribal) 2007-2008	526.2672
## Brazil (Sete De Setembro Reserve) 2003	1377.4105
## Pakistan (Karachi) 2002	431.0345
## Uganda (National) 2014-2015	686.3025
## Namibia (National) 2017-2018	563.7158
## Philippines (National) 2016	1541.1860
## Mongolia (National) 2014-2015	797.2098
## Myanmar (National) 2017-2018	843.3089
## Lesotho (National) 2019	1023.6129
## DPR Korea (National) 2015-2016	876.6738
## Bangladesh (National) 2015-2016	453.0598
## Indonesia (National) 2013-2014	913.6318
## Zimbabwe (National) 2014	408.5370
## South Africa (National) 2017-2019	926.2026
## China (Multiple Areas) 2013	247.0898
## India (Kashmir Valley) 2017	31.8558
## India (National Tribal) 2015-2020	377.1248
## India (Saharia Tribal) 2013-2013	4831.6252
## Viet Nam (National) 2017-2018	644.5672
## Ethiopia (Hawassa Zuria District) 2016	131.0401
## Ethiopia (Hetosa District) 2013	112.3795
## Malawi (Blantyre) 2019-2020	293.5900
## India (Tiruvallur District) 2015-2018	555.2170
## India (Pohari Block, Shivpuri District) 2012-2013, 2014-2015	4831.6252
## India (Saharia Tribal) 2019	2170.4500
##	95%-CI
## India (Gujarat State) 2011-2012	[482.7056; 624.5324]
## India (Wardha District) 2007-2009	[185.7393; 311.7239]
## India (Saharia Tribal) 2015-2015	[2687.9519; 3711.7372]
## South Africa (Western Cape Province, a) 2005	[1823.2251; 3507.2804]
## Viet Nam (National) 2006-2007	[433.0182; 569.0979]
## Malawi (National) 2013-2014	[370.0520; 615.4032]
## China (Yunnan Province) NA	[1280.9176; 2033.7938]
## Kenya (Asembo Area) 2006-2007	[580.3752; 986.1034]
## Phillipines (National) 2007	[790.4315; 1163.3657]
## India (Faridabad District) 2008-2009	[107.8091; 174.6729]
## Uganda (Rubaga) 2008-2009	[256.2292; 702.7582]
## India (Karachal) 2007-2008	[1753.3217; 2547.7217]
## India (Jabalpur District) 2009-2010	[302.3403; 410.5744]
## Pakistan (National) 2010-2011	[347.2873; 467.0618]
## Kenya (Karemo) NA	[134.0136; 610.2936]

```

## The Gambia (National) 2012-2013 [ 207.1678; 369.9234]
## Tanzania (National) 2011-2012 [ 349.1666; 533.2540]
## Nigeria (National) 2012 [ 378.5874; 583.9488]
## Myanmar (National) 2009-2010 [ 799.0212; 1053.7407]
## Cambodia (National) 2002 [1624.4670; 2242.8344]
## Lao Pdr (National) 2010-2011 [ 752.5502; 1034.1172]
## Viet Nam (Hanoi City) 2003-2004 [ 110.5464; 423.3314]
## Rwanda (National) 2012 [ 111.2709; 235.2941]
## India (Chennai) 2010-2012 [ 420.2668; 592.7895]
## India (Madhya Pradesh Tribal) 2007-2008 [ 398.8261; 681.3094]
## Brazil (Sete De Setembro Reserve) 2003 [ 448.7139; 3184.9809]
## Pakistan (Karachi) 2002 [ 215.3613; 769.9280]
## Uganda (National) 2014-2015 [ 569.3311; 820.0977]
## Namibia (National) 2017-2018 [ 440.5214; 710.5265]
## Philippines (National) 2016 [1378.5287; 1717.5217]
## Mongolia (National) 2014-2015 [ 678.8547; 930.1317]
## Myanmar (National) 2017-2018 [ 739.5133; 957.4947]
## Lesotho (National) 2019 [ 821.7541; 1259.6195]
## DPR Korea (National) 2015-2016 [ 769.0075; 995.0882]
## Bangladesh (National) 2015-2016 [ 392.7024; 520.0349]
## Indonesia (National) 2013-2014 [ 811.7261; 1024.6979]
## Zimbabwe (National) 2014 [ 310.3611; 527.8128]
## South Africa (National) 2017-2019 [ 770.9405; 1103.3197]
## China (Multiple Areas) 2013 [ 180.2844; 330.4875]
## India (Kashmir Valley) 2017 [ 12.8086; 65.6241]
## India (National Tribal) 2015-2020 [ 316.0701; 446.4972]
## India (Saharia Tribal) 2013-2013 [4241.0858; 5478.0717]
## Viet Nam (National) 2017-2018 [ 552.8479; 747.0760]
## Ethiopia (Hawassa Zuria District) 2016 [ 74.9187; 212.7140]
## Ethiopia (Hetosa District) 2013 [ 67.6729; 175.4391]
## Malawi (Blantyre) 2019-2020 [ 174.0900; 463.6033]
## India (Tiruvallur District) 2015-2018 [ 473.6788; 646.6984]
## India (Pohari Block, Shivpuri District) 2012-2013, 2014-2015 [4241.0858; 5478.0717]
## India (Saharia Tribal) 2019 [1894.4328; 2474.6612]
##
## Number of studies: k = 49
## Number of observations: o = 902032
## Number of events: e = 5711
##
##
## events 95%-CI
## Common effect model 633.1261 [616.9663; 649.7064]
## Random effects model 602.4039 [460.8503; 787.0931]
##
## Quantifying heterogeneity (with 95%-CIs):
## tau^2 = 0.8998; tau = 0.9486; I^2 = 98.7% [98.6%; 98.9%]; H = 8.89 [8.41; 9.40]
##
## Test of heterogeneity:
## Q d.f. p-value
## Wald 3796.32 48 0
## LRT 3553.04 48 0
##
## Details of meta-analysis methods:
## - Random intercept logistic regression model
## - Maximum-likelihood estimator for tau^2

```

```
## - Calculation of I2 based on Q
## - Logit transformation
## - Clopper-Pearson confidence interval for individual studies
## - Events per 100000 observations
```

Females

```
propBactFemale <-metaprop(n.bacteriological.tb.female, n.participants.female, figure.id.yr,
                          data=prevSexBact, random=TRUE,
                          pscale = 1e5)

print(summary(propBactFemale))
```

	events
## India (Gujarat State) 2011-2012	219.0803
## India (Wardha District) 2007-2009	99.1906
## India (Saharia Tribal) 2015-2015	906.4039
## South Africa (Western Cape Province, a) 2005	1994.1634
## Viet Nam (National) 2006-2007	110.5358
## Malawi (National) 2013-2014	373.1747
## China (Yunnan Province) NA	554.1347
## Kenya (Asembo Area) 2006-2007	501.1179
## Phillipines (National) 2007	374.0224
## India (Faridabad District) 2008-2009	62.5313
## Uganda (Rubaga) 2008-2009	1782.8201
## India (Karachal) 2007-2008	914.5815
## India (Jabalpur District) 2009-2010	108.5892
## Pakistan (National) 2010-2011	261.8701
## Kenya (Karemo) NA	329.8969
## The Gambia (National) 2012-2013	85.9509
## Tanzania (National) 2011-2012	215.4810
## Nigeria (National) 2012	223.0083
## Myanmar (National) 2009-2010	362.4064
## Cambodia (National) 2002	1188.7682
## Lao Pdr (National) 2010-2011	372.5435
## Viet Nam (Hanoi City) 2003-2004	185.2710
## Rwanda (National) 2012	40.1075
## India (Chennai) 2010-2012	122.0618
## India (Madhya Pradesh Tribal) 2007-2008	227.2926
## Brazil (Sete De Setembro Reserve) 2003	1340.4826
## Pakistan (Karachi) 2002	239.1527
## Uganda (National) 2014-2015	168.9974
## Namibia (National) 2017-2018	284.0237
## Philippines (National) 2016	558.2261
## Mongolia (National) 2014-2015	291.0149
## Myanmar (National) 2017-2018	223.4056
## Lesotho (National) 2019	335.3147
## DPR Korea (National) 2015-2016	306.1012
## Bangladesh (National) 2015-2016	141.6874
## Indonesia (National) 2013-2014	377.2857
## Zimbabwe (National) 2014	250.7548
## South Africa (National) 2017-2019	504.5177
## China (Multiple Areas) 2013	105.8728
## India (Kashmir Valley) 2017	81.5035
## India (National Tribal) 2015-2020	166.6667

## India (Saharia Tribal) 2013-2013	1246.2312	
## Viet Nam (National) 2017-2018	132.8980	
## Ethiopia (Hawassa Zuria District) 2016	146.2582	
## Ethiopia (Hetosa District) 2013	148.4597	
## Malawi (Blantyre) 2019-2020	116.7356	
## India (Tiruvallur District) 2015-2018	84.9618	
## India (Pohari Block, Shivpuri District) 2012-2013, 2014-2015	1246.2312	
## India (Saharia Tribal) 2019	546.1768	
##		95%-CI
## India (Gujarat State) 2011-2012	[177.6942;	267.1950]
## India (Wardha District) 2007-2009	[64.2009;	146.3903]
## India (Saharia Tribal) 2015-2015	[664.3441;	1207.1844]
## South Africa (Western Cape Province, a) 2005	[1434.7529;	2695.6653]
## Viet Nam (National) 2006-2007	[83.7292;	143.1886]
## Malawi (National) 2013-2014	[290.4647;	472.0423]
## China (Yunnan Province) NA	[362.2890;	810.8887]
## Kenya (Asembo Area) 2006-2007	[386.9588;	638.2761]
## Phillipines (National) 2007	[271.8928;	501.7860]
## India (Faridabad District) 2008-2009	[42.1934;	89.2553]
## Uganda (Rubaga) 2008-2009	[1120.5709;	2686.8290]
## India (Karachal) 2007-2008	[685.8136;	1194.6080]
## India (Jabalpur District) 2009-2010	[80.8621;	142.7503]
## Pakistan (National) 2010-2011	[222.9070;	305.6701]
## Kenya (Karemo) NA	[142.5303;	648.9883]
## The Gambia (National) 2012-2013	[53.8726;	130.1020]
## Tanzania (National) 2011-2012	[165.9849;	275.0822]
## Nigeria (National) 2012	[169.3818;	288.1958]
## Myanmar (National) 2009-2010	[296.5053;	438.5475]
## Cambodia (National) 2002	[983.2695;	1424.1207]
## Lao Pdr (National) 2010-2011	[295.5110;	463.4513]
## Viet Nam (Hanoi City) 2003-2004	[95.7677;	323.4072]
## Rwanda (National) 2012	[19.2347;	73.7467]
## India (Chennai) 2010-2012	[85.0348;	169.7179]
## India (Madhya Pradesh Tribal) 2007-2008	[148.5271;	332.8602]
## Brazil (Sete De Setembro Reserve) 2003	[436.6454;	3100.3584]
## Pakistan (Karachi) 2002	[96.2042;	492.1214]
## Uganda (National) 2014-2015	[120.7608;	230.0562]
## Namibia (National) 2017-2018	[209.4886;	376.3999]
## Philippines (National) 2016	[470.9732;	656.8829]
## Mongolia (National) 2014-2015	[233.4659;	358.4174]
## Myanmar (National) 2017-2018	[178.7333;	275.8317]
## Lesotho (National) 2019	[243.7427;	449.8858]
## DPR Korea (National) 2015-2016	[249.9164;	371.1158]
## Bangladesh (National) 2015-2016	[111.8331;	177.0536]
## Indonesia (National) 2013-2014	[316.8479;	445.8601]
## Zimbabwe (National) 2014	[185.5656;	331.3773]
## South Africa (National) 2017-2019	[414.8293;	607.7651]
## China (Multiple Areas) 2013	[61.6865;	169.4587]
## India (Kashmir Valley) 2017	[47.4857;	130.4630]
## India (National Tribal) 2015-2020	[128.6526;	212.3817]
## India (Saharia Tribal) 2013-2013	[956.7767;	1594.8001]
## Viet Nam (National) 2017-2018	[97.3140;	177.2280]
## Ethiopia (Hawassa Zuria District) 2016	[86.7042;	231.0529]
## Ethiopia (Hetosa District) 2013	[95.1434;	220.8163]

```
## Malawi (Blantyre) 2019-2020 [ 58.2880; 208.7761]
## India (Tiruvallur District) 2015-2018 [ 56.4637; 122.7702]
## India (Pohari Block, Shivpuri District) 2012-2013, 2014-2015 [ 956.7767; 1594.8001]
## India (Saharia Tribal) 2019 [ 411.7123; 710.3381]
```

```
##
## Number of studies: k = 49
## Number of observations: o = 1073194
## Number of events: e = 2665
##
##              events          95%-CI
## Common effect model 248.3242 [239.0838; 257.9208]
## Random effects model 271.1049 [210.8979; 348.4398]
##
## Quantifying heterogeneity (with 95%-CIs):
## tau^2 = 0.7730; tau = 0.8792; I^2 = 96.9% [96.4%; 97.3%]; H = 5.68 [5.27; 6.11]
##
## Test of heterogeneity:
##      Q d.f.  p-value
## Wald 1547.25  48 < 0.0001
## LRT 1455.93  48 < 0.0001
##
## Details of meta-analysis methods:
## - Random intercept logistic regression model
## - Maximum-likelihood estimator for tau^2
## - Calculation of I^2 based on Q
## - Logit transformation
## - Clopper-Pearson confidence interval for individual studies
## - Events per 100000 observations
```

```
### Option 2: Use those surveys which report counts of participants
### and bacteriological TB AND those that report prev100k.
```

```
prevSexBact2 <- cleanSexDFO %>%
  mutate(n.bacteriological.tb.male = ifelse(is.na(n.bacteriological.tb.male) &
    !is.na(prev100k.bacteriological.tb.male),
    round(prev100k.bacteriological.tb.male*n.participants.male/
    n.bacteriological.tb.male),
    n.bacteriological.tb.female = ifelse(is.na(n.bacteriological.tb.female) &
    !is.na(prev100k.bacteriological.tb.female),
    round(prev100k.bacteriological.tb.female*n.participants.female/
    n.bacteriological.tb.female)) %>%
  filter(is.na(n.bacteriological.tb.male)== FALSE &
    is.na(n.participants.male)== FALSE)

### Males
propBactMale2 <-metaprop(n.bacteriological.tb.male, n.participants.male, figure.id.yr,
  data=prevSexBact2, random=TRUE,
  pscale = 1e5)

summary(propBactMale2)
```

```
##
##              events
## India (Gujarat State) 2011-2012 550.2381
## India (Wardha District) 2007-2009 242.7571
```

## India (Saharia Tribal) 2015-2015	3170.2128
## South Africa (Western Cape Province, a) 2005	2567.5676
## Viet Nam (National) 2006-2007	497.6175
## Malawi (National) 2013-2014	481.3202
## China (Yunnan Province) NA	1625.8400
## Kenya (Asembo Area) 2006-2007	763.6603
## Phillipines (National) 2007	963.7035
## India (Faridabad District) 2008-2009	138.2771
## Uganda (Rubaga) 2008-2009	439.5036
## India (Karachal) 2007-2008	2123.6610
## India (Jabalpur District) 2009-2010	353.3936
## Pakistan (National) 2010-2011	403.8916
## Kenya (Karemo) NA	310.1978
## The Gambia (National) 2012-2013	279.9360
## Tanzania (National) 2011-2012	434.0487
## Nigeria (National) 2012	473.0994
## Myanmar (National) 2009-2010	919.8893
## Cambodia (National) 2002	1915.5144
## Lao Pdr (National) 2010-2011	885.1054
## Viet Nam (Hanoi City) 2003-2004	230.4147
## Rwanda (National) 2012	164.8805
## India (Chennai) 2010-2012	501.0578
## Zimbabwe (Harare) 2005-2006	780.8564
## India (Nelamangala Sub-Division) 2008-2010	315.8284
## India (Madhya Pradesh Tribal) 2007-2008	526.2672
## Brazil (Sete De Setembro Reserve) 2003	1377.4105
## Pakistan (Karachi) 2002	431.0345
## India (Mohali District) 2008-2010	34.4693
## Uganda (National) 2014-2015	686.3025
## Namibia (National) 2017-2018	563.7158
## Philippines (National) 2016	1541.1860
## Mongolia (National) 2014-2015	797.2098
## Eswatini (National) 2018-2019	342.1212
## Myanmar (National) 2017-2018	843.3089
## India (National) 2019-2021	496.1085
## Lesotho (National) 2019	1023.6129
## DPR Korea (National) 2015-2016	876.6738
## Kenya (National) 2015-2016	529.8725
## Bangladesh (National) 2015-2016	453.0598
## Indonesia (National) 2013-2014	913.6318
## Zimbabwe (National) 2014	408.5370
## South Africa (National) 2017-2019	926.2026
## China (Multiple Areas) 2013	247.0898
## India (Kashmir Valley) 2017	31.8558
## India (National Tribal) 2015-2020	377.1248
## India (Saharia Tribal) 2013-2013	4831.6252
## Viet Nam (National) 2017-2018	644.5672
## China (Xinjiang Uyghur Region) 2010-2011	429.4224
## Ethiopia (Hawassa Zuria District) 2016	131.0401
## Ethiopia (Hetosa District) 2013	112.3795
## Malawi (Blantyre) 2019-2020	293.5900
## India (Tiruvallur District) 2015-2018	555.2170
## India (Pohari Block, Shivpuri District) 2012-2013, 2014-2015	4831.6252
## India (Saharia Tribal) 2019	2170.4500

	95%-CI
##	
## India (Gujarat State) 2011-2012	[482.7056; 624.5324]
## India (Wardha District) 2007-2009	[185.7393; 311.7239]
## India (Saharia Tribal) 2015-2015	[2687.9519; 3711.7372]
## South Africa (Western Cape Province, a) 2005	[1823.2251; 3507.2804]
## Viet Nam (National) 2006-2007	[433.0182; 569.0979]
## Malawi (National) 2013-2014	[370.0520; 615.4032]
## China (Yunnan Province) NA	[1280.9176; 2033.7938]
## Kenya (Asembo Area) 2006-2007	[580.3752; 986.1034]
## Phillipines (National) 2007	[790.4315; 1163.3657]
## India (Faridabad District) 2008-2009	[107.8091; 174.6729]
## Uganda (Rubaga) 2008-2009	[256.2292; 702.7582]
## India (Karachal) 2007-2008	[1753.3217; 2547.7217]
## India (Jabalpur District) 2009-2010	[302.3403; 410.5744]
## Pakistan (National) 2010-2011	[347.2873; 467.0618]
## Kenya (Karemo) NA	[134.0136; 610.2936]
## The Gambia (National) 2012-2013	[207.1678; 369.9234]
## Tanzania (National) 2011-2012	[349.1666; 533.2540]
## Nigeria (National) 2012	[378.5874; 583.9488]
## Myanmar (National) 2009-2010	[799.0212; 1053.7407]
## Cambodia (National) 2002	[1624.4670; 2242.8344]
## Lao Pdr (National) 2010-2011	[752.5502; 1034.1172]
## Viet Nam (Hanoi City) 2003-2004	[110.5464; 423.3314]
## Rwanda (National) 2012	[111.2709; 235.2941]
## India (Chennai) 2010-2012	[420.2668; 592.7895]
## Zimbabwe (Harare) 2005-2006	[531.1529; 1106.5470]
## India (Nelamangala Sub-Division) 2008-2010	[255.2946; 386.3573]
## India (Madhya Pradesh Tribal) 2007-2008	[398.8261; 681.3094]
## Brazil (Sete De Setembro Reserve) 2003	[448.7139; 3184.9809]
## Pakistan (Karachi) 2002	[215.3613; 769.9280]
## India (Mohali District) 2008-2010	[19.2934; 56.8455]
## Uganda (National) 2014-2015	[569.3311; 820.0977]
## Namibia (National) 2017-2018	[440.5214; 710.5265]
## Philippines (National) 2016	[1378.5287; 1717.5217]
## Mongolia (National) 2014-2015	[678.8547; 930.1317]
## Eswatini (National) 2018-2019	[237.0417; 477.7549]
## Myanmar (National) 2017-2018	[739.5133; 957.4947]
## India (National) 2019-2021	[460.2397; 534.0195]
## Lesotho (National) 2019	[821.7541; 1259.6195]
## DPR Korea (National) 2015-2016	[769.0075; 995.0882]
## Kenya (National) 2015-2016	[445.3382; 625.7152]
## Bangladesh (National) 2015-2016	[392.7024; 520.0349]
## Indonesia (National) 2013-2014	[811.7261; 1024.6979]
## Zimbabwe (National) 2014	[310.3611; 527.8128]
## South Africa (National) 2017-2019	[770.9405; 1103.3197]
## China (Multiple Areas) 2013	[180.2844; 330.4875]
## India (Kashmir Valley) 2017	[12.8086; 65.6241]
## India (National Tribal) 2015-2020	[316.0701; 446.4972]
## India (Saharia Tribal) 2013-2013	[4241.0858; 5478.0717]
## Viet Nam (National) 2017-2018	[552.8479; 747.0760]
## China (Xinjiang Uyghur Region) 2010-2011	[329.3894; 550.1666]
## Ethiopia (Hawassa Zuria District) 2016	[74.9187; 212.7140]
## Ethiopia (Hetosa District) 2013	[67.6729; 175.4391]
## Malawi (Blantyre) 2019-2020	[174.0900; 463.6033]

```

## India (Tiruvallur District) 2015-2018 [ 473.6788; 646.6984]
## India (Pohari Block, Shivpuri District) 2012-2013, 2014-2015 [4241.0858; 5478.0717]
## India (Saharia Tribal) 2019 [1894.4328; 2474.6612]
##
## Number of studies: k = 56
## Number of observations: o = 1171808
## Number of events: e = 6790
##
##              events          95%-CI
## Common effect model 579.4465 [565.8648; 593.3522]
## Random effects model 556.1562 [430.6964; 717.8982]
##
## Quantifying heterogeneity (with 95%-CIs):
## tau^2 = 0.9372; tau = 0.9681; I^2 = 98.7% [98.6%; 98.8%]; H = 8.82 [8.38; 9.29]
##
## Test of heterogeneity:
##           Q d.f. p-value
## Wald 4282.44  55      0
## LRT 4070.10  55      0
##
## Details of meta-analysis methods:
## - Random intercept logistic regression model
## - Maximum-likelihood estimator for tau^2
## - Calculation of I^2 based on Q
## - Logit transformation
## - Clopper-Pearson confidence interval for individual studies
## - Events per 100000 observations

```

Females

```

propBactFemale2 <-metaprop(n.bacteriological.tb.female, n.participants.female, figure.id.yr,
                          data=prevSexBact2, random=TRUE,
                          pscale = 1e5)

summary(propBactFemale2)

```

```

##              events
## India (Gujarat State) 2011-2012 219.0803
## India (Wardha District) 2007-2009 99.1906
## India (Saharia Tribal) 2015-2015 906.4039
## South Africa (Western Cape Province, a) 2005 1994.1634
## Viet Nam (National) 2006-2007 110.5358
## Malawi (National) 2013-2014 373.1747
## China (Yunnan Province) NA 554.1347
## Kenya (Asembo Area) 2006-2007 501.1179
## Phillipines (National) 2007 374.0224
## India (Faridabad District) 2008-2009 62.5313
## Uganda (Rubaga) 2008-2009 1782.8201
## India (Karachal) 2007-2008 914.5815
## India (Jabalpur District) 2009-2010 108.5892
## Pakistan (National) 2010-2011 261.8701
## Kenya (Karemo) NA 329.8969
## The Gambia (National) 2012-2013 85.9509
## Tanzania (National) 2011-2012 215.4810
## Nigeria (National) 2012 223.0083

```

## Myanmar (National) 2009-2010	362.4064
## Cambodia (National) 2002	1188.7682
## Lao Pdr (National) 2010-2011	372.5435
## Viet Nam (Hanoi City) 2003-2004	185.2710
## Rwanda (National) 2012	40.1075
## India (Chennai) 2010-2012	122.0618
## Zimbabwe (Harare) 2005-2006	571.8020
## India (Nelamangala Sub-Division) 2008-2010	50.5967
## India (Madhya Pradesh Tribal) 2007-2008	227.2926
## Brazil (Sete De Setembro Reserve) 2003	1340.4826
## Pakistan (Karachi) 2002	239.1527
## India (Mohali District) 2008-2010	14.2002
## Uganda (National) 2014-2015	168.9974
## Namibia (National) 2017-2018	284.0237
## Philippines (National) 2016	558.2261
## Mongolia (National) 2014-2015	291.0149
## Eswatini (National) 2018-2019	242.7521
## Myanmar (National) 2017-2018	223.4056
## India (National) 2019-2021	153.0307
## Lesotho (National) 2019	335.3147
## DPR Korea (National) 2015-2016	306.1012
## Kenya (National) 2015-2016	208.0744
## Bangladesh (National) 2015-2016	141.6874
## Indonesia (National) 2013-2014	377.2857
## Zimbabwe (National) 2014	250.7548
## South Africa (National) 2017-2019	504.5177
## China (Multiple Areas) 2013	105.8728
## India (Kashmir Valley) 2017	81.5035
## India (National Tribal) 2015-2020	166.6667
## India (Saharia Tribal) 2013-2013	1246.2312
## Viet Nam (National) 2017-2018	132.8980
## China (Xinjiang Uyghur Region) 2010-2011	259.7909
## Ethiopia (Hawassa Zuria District) 2016	146.2582
## Ethiopia (Hetosa District) 2013	148.4597
## Malawi (Blantyre) 2019-2020	116.7356
## India (Tiruvallur District) 2015-2018	84.9618
## India (Pohari Block, Shivpuri District) 2012-2013, 2014-2015	1246.2312
## India (Saharia Tribal) 2019	546.1768
##	95%-CI
## India (Gujarat State) 2011-2012	[177.6942; 267.1950]
## India (Wardha District) 2007-2009	[64.2009; 146.3903]
## India (Saharia Tribal) 2015-2015	[664.3441; 1207.1844]
## South Africa (Western Cape Province, a) 2005	[1434.7529; 2695.6653]
## Viet Nam (National) 2006-2007	[83.7292; 143.1886]
## Malawi (National) 2013-2014	[290.4647; 472.0423]
## China (Yunnan Province) NA	[362.2890; 810.8887]
## Kenya (Asembo Area) 2006-2007	[386.9588; 638.2761]
## Phillipines (National) 2007	[271.8928; 501.7860]
## India (Faridabad District) 2008-2009	[42.1934; 89.2553]
## Uganda (Rubaga) 2008-2009	[1120.5709; 2686.8290]
## India (Karachal) 2007-2008	[685.8136; 1194.6080]
## India (Jabalpur District) 2009-2010	[80.8621; 142.7503]
## Pakistan (National) 2010-2011	[222.9070; 305.6701]
## Kenya (Karemo) NA	[142.5303; 648.9883]

```

## The Gambia (National) 2012-2013 [ 53.8726; 130.1020]
## Tanzania (National) 2011-2012 [ 165.9849; 275.0822]
## Nigeria (National) 2012 [ 169.3818; 288.1958]
## Myanmar (National) 2009-2010 [ 296.5053; 438.5475]
## Cambodia (National) 2002 [ 983.2695; 1424.1207]
## Lao Pdr (National) 2010-2011 [ 295.5110; 463.4513]
## Viet Nam (Hanoi City) 2003-2004 [ 95.7677; 323.4072]
## Rwanda (National) 2012 [ 19.2347; 73.7467]
## India (Chennai) 2010-2012 [ 85.0348; 169.7179]
## Zimbabwe (Harare) 2005-2006 [ 398.5947; 794.3492]
## India (Nelamangala Sub-Division) 2008-2010 [ 29.4771; 80.9980]
## India (Madhya Pradesh Tribal) 2007-2008 [ 148.5271; 332.8602]
## Brazil (Sete De Setembro Reserve) 2003 [ 436.6454; 3100.3584]
## Pakistan (Karachi) 2002 [ 96.2042; 492.1214]
## India (Mohali District) 2008-2010 [ 5.2114; 30.9052]
## Uganda (National) 2014-2015 [ 120.7608; 230.0562]
## Namibia (National) 2017-2018 [ 209.4886; 376.3999]
## Philippines (National) 2016 [ 470.9732; 656.8829]
## Mongolia (National) 2014-2015 [ 233.4659; 358.4174]
## Eswatini (National) 2018-2019 [ 169.1422; 337.4493]
## Myanmar (National) 2017-2018 [ 178.7333; 275.8317]
## India (National) 2019-2021 [ 135.5190; 172.1745]
## Lesotho (National) 2019 [ 243.7427; 449.8858]
## DPR Korea (National) 2015-2016 [ 249.9164; 371.1158]
## Kenya (National) 2015-2016 [ 164.2429; 259.9896]
## Bangladesh (National) 2015-2016 [ 111.8331; 177.0536]
## Indonesia (National) 2013-2014 [ 316.8479; 445.8601]
## Zimbabwe (National) 2014 [ 185.5656; 331.3773]
## South Africa (National) 2017-2019 [ 414.8293; 607.7651]
## China (Multiple Areas) 2013 [ 61.6865; 169.4587]
## India (Kashmir Valley) 2017 [ 47.4857; 130.4630]
## India (National Tribal) 2015-2020 [ 128.6526; 212.3817]
## India (Saharia Tribal) 2013-2013 [ 956.7767; 1594.8001]
## Viet Nam (National) 2017-2018 [ 97.3140; 177.2280]
## China (Xinjiang Uyghur Region) 2010-2011 [ 185.6613; 353.5952]
## Ethiopia (Hawassa Zuria District) 2016 [ 86.7042; 231.0529]
## Ethiopia (Hetosa District) 2013 [ 95.1434; 220.8163]
## Malawi (Blantyre) 2019-2020 [ 58.2880; 208.7761]
## India (Tiruvallur District) 2015-2018 [ 56.4637; 122.7702]
## India (Pohari Block, Shivpuri District) 2012-2013, 2014-2015 [ 956.7767; 1594.8001]
## India (Saharia Tribal) 2019 [ 411.7123; 710.3381]
##
## Number of studies: k = 56
## Number of observations: o = 1402344
## Number of events: e = 3151
##
## events 95%-CI
## Common effect model 224.6952 [216.9934; 232.6698]
## Random effects model 249.3051 [194.5155; 319.4780]
##
## Quantifying heterogeneity (with 95%-CIs):
## tau^2 = 0.8657; tau = 0.9304; I^2 = 97.0% [96.5%; 97.4%]; H = 5.76 [5.38; 6.17]
##
## Test of heterogeneity:

```

```

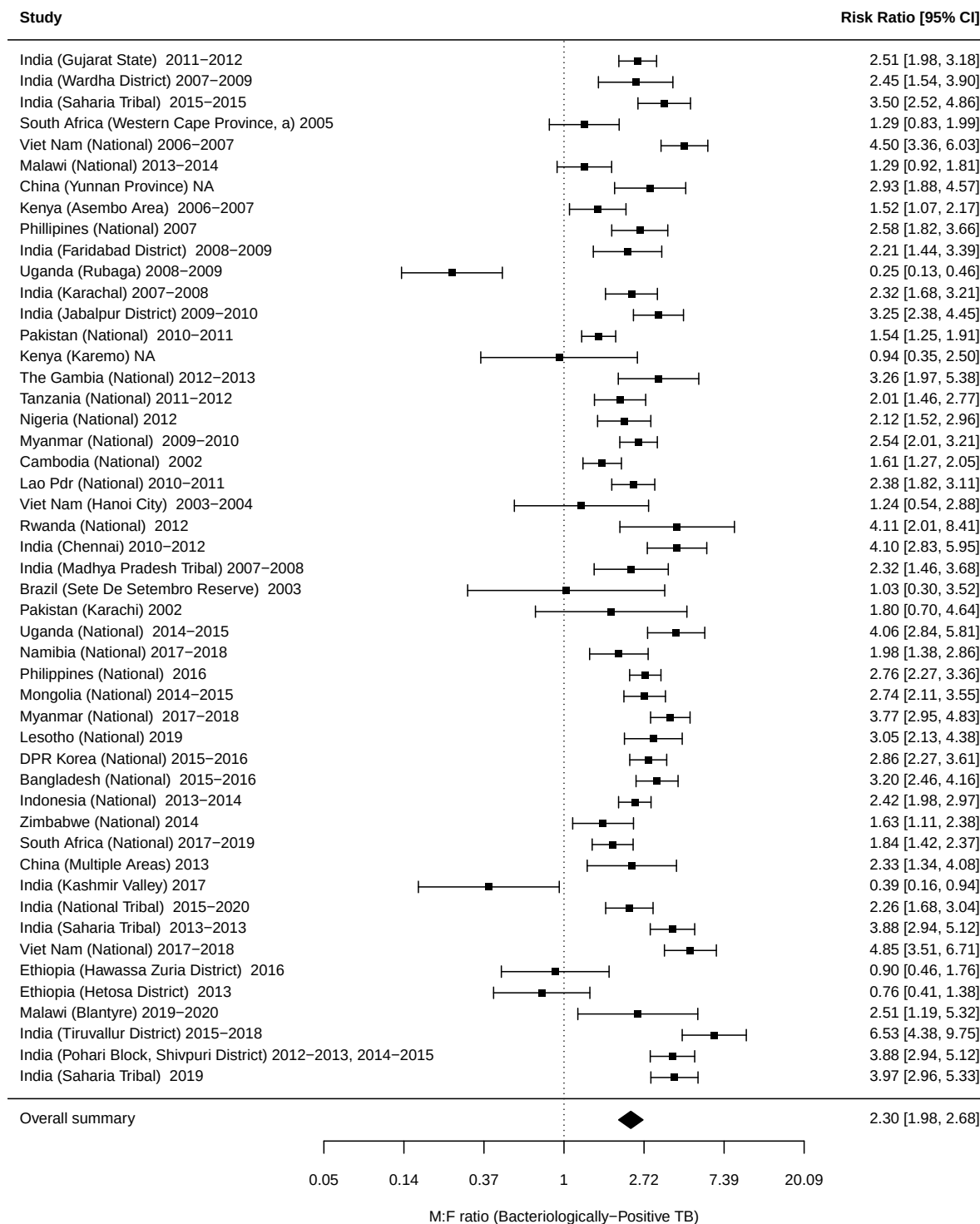
##           Q d.f. p-value
## Wald 1827.28  55      0
## LRT  1764.11  55      0
##
## Details of meta-analysis methods:
## - Random intercept logistic regression model
## - Maximum-likelihood estimator for tau^2
## - Calculation of I^2 based on Q
## - Logit transformation
## - Clopper-Pearson confidence interval for individual studies
## - Events per 100000 observations

#####|
##### MALE TO FEMALE RATIO OF BACTERIOLOGICAL PREVALENCE #####|
#####|

bactPrevMF <- rma(ai=n.bacteriological.tb.male,
                 bi=n.participants.male - n.bacteriological.tb.male,
                 ci=n.bacteriological.tb.female,
                 di=n.participants.female - n.bacteriological.tb.female,
                 data=prevSexBact, measure="RR",
                 slab=paste(figure.id.yr), method="REML")

forest(bactPrevMF,
       atransf=exp, cex=.75, xlab="M:F ratio (Bacteriologically-Positive TB)", mlab="Overall summary",
       psize=1, col="black", border="black" )

```



 ##### SEX STRATIFIED SMEAR POSITIVE TB PREVALENCE #####

```
#####/

### Calculation of smear positive TB prevalence among males
### Random effects weighted model
prevSexSmear <- cleanSexDF0 %>% filter(is.na(n.smear.positive.tb.male)== FALSE &
                                         is.na(n.participants.male)== FALSE )

propSmearMale <-metaprop(n.smear.positive.tb.male, n.participants.male, figure.id.yr,
                        data=prevSexSmear, random=TRUE, pscale = 1e5)

summary(propSmearMale)
```

##	events
## India (Tiruvallur District) 2001-2003	396.5679
## India (Tiruvallur District) 2006-2008	282.2473
## India (Tiruvallur District) 2004-2006	250.7073
## India (Gujarat State) 2011-2012	386.0915
## India (Saharia Tribal) 2015-2015	2446.8085
## India (Assam State) 2008-2010	1420.9591
## Viet Nam (National) 2006-2007	333.3099
## Ethiopia (Shebedino District) 2012-2013	67.3627
## Ethiopia (Lemu District) 2012-2013	237.4429
## Malawi (National) 2013-2014	595.9202
## Bangladesh (Matlab) 2001	165.0100
## Bangladesh (National) 2007-2009	99.1613
## Viet Nam (Central Highlands) 2003	25.8554
## Kenya (Asembo Area) 2006-2007	302.8308
## Phillipines (National) 2007	342.2498
## Viet Nam (Bavi District) 2000	59.7479
## Ethiopia (Dabat District) 2010	80.4290
## Indonesia (National) 2004	209.9889
## Uganda (Kisenyi) 2005	2560.0000
## Eritrea (National) 2005	113.1405
## India (Andhra Pradesh State) 2008-2010	436.6812
## India (Jabalpur District) 2009-2010	239.0604
## Pakistan (National) 2010-2011	321.3282
## The Gambia (National) 2012-2013	142.8245
## Tanzania (National) 2011-2012	385.8211
## Nigeria (National) 2012	363.0762
## Myanmar (National) 1994	239.5954
## Myanmar (National) 2009-2010	392.9624
## Ethiopia (National) 2010-2011	119.1622
## Cambodia (National) 2002	697.7039
## Lao Pdr (National) 2010-2011	422.8211
## Viet Nam (Hanoi City) 2003-2004	161.2903
## Bangladesh (Multiple Areas) 2001	46.3710
## India (R.S. Pura Block, Jammu City) Not reported	297.2873
## India (Tiruvallur District) 1999-2001	575.2050
## Rwanda (National) 2012	126.4084
## India (Chennai) 2010-2012	337.7501
## Ethiopia (District 16, Addis Ababa) 2001	228.3539
## Zimbabwe (Harare) NA	571.8548
## Brazil (Sete De Setembro Reserve) 2003	826.4463

## Pakistan (Shimshal Valley) 1996	1020.4082	
## Uganda (National) 2014-2015	291.6786	
## Philippines (National) 2016	598.2865	
## Mongolia (National) 2014-2015	338.8142	
## Indonesia (National) 2013-2014	379.3627	
## Zimbabwe (National) 2014	91.5686	
## India (National Tribal) 2015-2020	309.5801	
## India (Saharia Tribal) 2013-2013	3262.9157	
## India (Maharashtra Tribal) 2015-2017	90.9918	
## Viet Nam (National) 2017-2018	125.2302	
## Ethiopia (Hetosa District) 2013	94.6354	
## India (Tiruvallur District) 2015-2018	240.3683	
## Ethiopia (Multiple Areas) 2012-2013	157.3564	
##		95%-CI
## India (Tiruvallur District) 2001-2003	[338.4605;	461.7563]
## India (Tiruvallur District) 2006-2008	[235.3495;	335.7307]
## India (Tiruvallur District) 2004-2006	[205.9006;	302.3493]
## India (Gujarat State) 2011-2012	[329.8430;	449.1517]
## India (Saharia Tribal) 2015-2015	[2024.2181;	2929.7928]
## India (Assam State) 2008-2010	[615.4126;	2780.5783]
## Viet Nam (National) 2006-2007	[280.8147;	392.7428]
## Ethiopia (Shebedino District) 2012-2013	[24.7248;	146.5622]
## Ethiopia (Lemu District) 2012-2013	[126.4870;	405.6924]
## Malawi (National) 2013-2014	[471.3269;	743.1852]
## Bangladesh (Matlab) 2001	[118.9498;	222.9809]
## Bangladesh (National) 2007-2009	[63.5445;	147.5084]
## Viet Nam (Central Highlands) 2003	[11.8234;	49.0759]
## Kenya (Asembo Area) 2006-2007	[192.0626;	454.0505]
## Phillipines (National) 2007	[242.3067;	469.4650]
## Viet Nam (Bavi District) 2000	[28.6550;	109.8509]
## Ethiopia (Dabat District) 2010	[36.7836;	152.6241]
## Indonesia (National) 2004	[156.3893;	276.0053]
## Uganda (Kisenyi) 2005	[1470.1828;	4124.0552]
## Eritrea (National) 2005	[45.5000;	232.9728]
## India (Andhra Pradesh State) 2008-2010	[160.4178;	948.0312]
## India (Jabalpur District) 2009-2010	[197.4080;	286.8870]
## Pakistan (National) 2010-2011	[271.0552;	378.1970]
## The Gambia (National) 2012-2013	[92.4491;	210.7654]
## Tanzania (National) 2011-2012	[306.0471;	479.9610]
## Nigeria (National) 2012	[280.9111;	461.6936]
## Myanmar (National) 1994	[157.9524;	348.4086]
## Myanmar (National) 2009-2010	[315.2830;	483.9198]
## Ethiopia (National) 2010-2011	[77.8550;	174.5518]
## Cambodia (National) 2002	[526.0260;	907.2010]
## Lao Pdr (National) 2010-2011	[332.7165;	529.7256]
## Viet Nam (Hanoi City) 2003-2004	[64.8709;	332.0356]
## Bangladesh (Multiple Areas) 2001	[34.1922;	61.4766]
## India (R.S. Pura Block, Jammu City) Not reported	[128.4322;	584.9297]
## India (Tiruvallur District) 1999-2001	[504.1814;	653.3848]
## Rwanda (National) 2012	[80.1484;	189.6146]
## India (Chennai) 2010-2012	[272.0198;	414.5226]
## Ethiopia (District 16, Addis Ababa) 2001	[118.0477;	398.5483]
## Zimbabwe (Harare) NA	[362.8420;	856.8346]
## Brazil (Sete De Setembro Reserve) 2003	[170.7584;	2396.1068]

```

## Pakistan (Shimshal Valley) 1996 [ 25.8312; 5554.4972]
## Uganda (National) 2014-2015 [ 217.2486; 383.3273]
## Philippines (National) 2016 [ 498.2466; 712.4227]
## Mongolia (National) 2014-2015 [ 263.1956; 429.3327]
## Indonesia (National) 2013-2014 [ 314.6265; 453.4562]
## Zimbabwe (National) 2014 [ 48.7652; 156.5343]
## India (National Tribal) 2015-2020 [ 254.5038; 373.0091]
## India (Saharia Tribal) 2013-2013 [2777.5711; 3806.3703]
## India (Maharashtra Tribal) 2015-2017 [ 18.7686; 265.6842]
## Viet Nam (National) 2017-2018 [ 86.7407; 174.9531]
## Ethiopia (Hetosa District) 2013 [ 54.1017; 153.6365]
## India (Tiruvallur District) 2015-2018 [ 187.7759; 303.0967]
## Ethiopia (Multiple Areas) 2012-2013 [ 67.9590; 309.8182]
##
## Number of studies: k = 53
## Number of observations: o = 1075514
## Number of events: e = 3092
##
##              events          95%-CI
## Common effect model 287.4904 [277.5472; 297.7888]
## Random effects model 267.7279 [206.8574; 346.4481]
##
## Quantifying heterogeneity (with 95%-CIs):
## tau^2 = 0.8565; tau = 0.9255; I^2 = 97.5% [97.1%; 97.8%]; H = 6.28 [5.88; 6.72]
##
## Test of heterogeneity:
##      Q d.f. p-value
## Wald 2053.52 52      0
## LRT 1922.21 52      0
##
## Details of meta-analysis methods:
## - Random intercept logistic regression model
## - Maximum-likelihood estimator for tau^2
## - Calculation of I^2 based on Q
## - Logit transformation
## - Clopper-Pearson confidence interval for individual studies
## - Events per 100000 observations

### Calculation of smear positive TB prevalence among females
### Random effects weighted model
propSmearFemale <-metaprop(n.smear.positive.tb.female, n.participants.female, figure.id.yr,
                          data=prevSexSmear, random=TRUE, pscale = 1e5)

summary(propSmearFemale)

##              events
## India (Tiruvallur District) 2001-2003 70.6682
## India (Tiruvallur District) 2006-2008 59.2480
## India (Tiruvallur District) 2004-2006 58.7774
## India (Gujarat State) 2011-2012 151.3235
## India (Saharia Tribal) 2015-2015 650.2463
## India (Assam State) 2008-2010 590.3188
## Viet Nam (National) 2006-2007 62.0552
## Ethiopia (Shebedino District) 2012-2013 129.4498

```

## Ethiopia (Lemu District) 2012-2013	56.3486	
## Malawi (National) 2013-2014	459.7080	
## Bangladesh (Matlab) 2001	29.4620	
## Bangladesh (National) 2007-2009	32.2638	
## Viet Nam (Central Highlands) 2003	17.5773	
## Kenya (Asembo Area) 2006-2007	215.8662	
## Phillipines (National) 2007	187.0112	
## Viet Nam (Bavi District) 2000	78.5546	
## Ethiopia (Dabat District) 2010	258.0645	
## Indonesia (National) 2004	112.1120	
## Uganda (Kisenyi) 2005	5573.7705	
## Eritrea (National) 2005	66.8617	
## India (Andhra Pradesh State) 2008-2010	315.0599	
## India (Jabalpur District) 2009-2010	70.2636	
## Pakistan (National) 2010-2011	162.0321	
## The Gambia (National) 2012-2013	42.9755	
## Tanzania (National) 2011-2012	164.9776	
## Nigeria (National) 2012	157.6438	
## Myanmar (National) 1994	79.0855	
## Myanmar (National) 2009-2010	120.8021	
## Ethiopia (National) 2010-2011	84.4119	
## Cambodia (National) 2002	235.7040	
## Lao Pdr (National) 2010-2011	149.0174	
## Viet Nam (Hanoi City) 2003-2004	154.3925	
## Bangladesh (Multiple Areas) 2001	13.2865	
## India (R.S. Pura Block, Jammu City) Not reported	86.6176	
## India (Tiruvallur District) 1999-2001	105.7952	
## Rwanda (National) 2012	16.0430	
## India (Chennai) 2010-2012	69.7496	
## Ethiopia (District 16, Addis Ababa) 2001	159.5590	
## Zimbabwe (Harare) NA	273.6198	
## Brazil (Sete De Setembro Reserve) 2003	804.2895	
## Pakistan (Shimshal Valley) 1996	3759.3985	
## Uganda (National) 2014-2015	63.3740	
## Philippines (National) 2016	186.0754	
## Mongolia (National) 2014-2015	66.1398	
## Indonesia (National) 2013-2014	123.9260	
## Zimbabwe (National) 2014	51.1745	
## India (National Tribal) 2015-2020	135.8974	
## India (Saharia Tribal) 2013-2013	743.7186	
## India (Maharashtra Tribal) 2015-2017	27.7701	
## Viet Nam (National) 2017-2018	46.2254	
## Ethiopia (Hetosa District) 2013	123.7164	
## India (Tiruvallur District) 2015-2018	30.3435	
## Ethiopia (Multiple Areas) 2012-2013	120.6030	
##		95%-CI
## India (Tiruvallur District) 2001-2003	[48.0205; 100.2929]	
## India (Tiruvallur District) 2006-2008	[39.3733; 85.6185]	
## India (Tiruvallur District) 2004-2006	[38.7382; 85.5066]	
## India (Gujarat State) 2011-2012	[117.2923; 192.1363]	
## India (Saharia Tribal) 2015-2015	[448.0100; 911.9862]	
## India (Assam State) 2008-2010	[191.9445; 1372.1912]	
## Viet Nam (National) 2006-2007	[42.4495; 87.5922]	
## Ethiopia (Shebedino District) 2012-2013	[55.9033; 254.9076]	

```

## Ethiopia (Lemu District) 2012-2013      [ 11.6219; 164.5854]
## Malawi (National) 2013-2014             [ 367.3585; 568.1267]
## Bangladesh (Matlab) 2001                [ 14.1291; 54.1750]
## Bangladesh (National) 2007-2009         [ 14.7541; 61.2380]
## Viet Nam (Central Highlands) 2003       [ 6.4508; 38.2544]
## Kenya (Asembo Area) 2006-2007         [ 143.4879; 311.8367]
## Phillipines (National) 2007             [ 117.2349; 283.0011]
## Viet Nam (Bavi District) 2000           [ 43.9729; 129.5308]
## Ethiopia (Dabat District) 2010          [ 176.5809; 364.1165]
## Indonesia (National) 2004              [ 75.0956; 160.9721]
## Uganda (Kisenyi) 2005                  [3280.0885; 8774.3454]
## Eritrea (National) 2005                [ 28.8704; 131.7014]
## India (Andhra Pradesh State) 2008-2010   [ 102.3758; 733.7010]
## India (Jabalpur District) 2009-2010     [ 48.3710; 98.6622]
## Pakistan (National) 2010-2011          [ 131.7106; 197.2335]
## The Gambia (National) 2012-2013        [ 21.4551; 76.8819]
## Tanzania (National) 2011-2012          [ 122.0755; 218.0512]
## Nigeria (National) 2012                [ 113.1509; 213.8015]
## Myanmar (National) 1994                [ 39.4856; 141.4616]
## Myanmar (National) 2009-2010           [ 84.1571; 167.9669]
## Ethiopia (National) 2010-2011          [ 52.2597; 129.0038]
## Cambodia (National) 2002               [ 149.4730; 353.4631]
## Lao Pdr (National) 2010-2011           [ 101.9494; 210.3036]
## Viet Nam (Hanoi City) 2003-2004         [ 74.0612; 283.7494]
## Bangladesh (Multiple Areas) 2001       [ 7.5946; 21.5755]
## India (R.S. Pura Block, Jammu City) Not reported [ 10.4915; 312.5387]
## India (Tiruvallur District) 1999-2001   [ 77.1779; 141.5370]
## Rwanda (National) 2012                 [ 4.3713; 41.0713]
## India (Chennai) 2010-2012              [ 42.6099; 107.7021]
## Ethiopia (District 16, Addis Ababa) 2001 [ 79.6774; 285.3154]
## Zimbabwe (Harare) NA                   [ 159.4719; 437.7316]
## Brazil (Sete De Setembro Reserve) 2003  [ 166.1720; 2332.3652]
## Pakistan (Shimshal Valley) 1996         [1231.7455; 8555.0407]
## Uganda (National) 2014-2015            [ 35.4742; 104.5043]
## Philippines (National) 2016            [ 137.2282; 246.6341]
## Mongolia (National) 2014-2015          [ 40.4044; 102.1291]
## Indonesia (National) 2013-2014         [ 90.4064; 165.7880]
## Zimbabwe (National) 2014               [ 24.5428; 94.0914]
## India (National Tribal) 2015-2020       [ 101.8126; 177.7200]
## India (Saharia Tribal) 2013-2013        [ 524.1715; 1023.6746]
## India (Maharashtra Tribal) 2015-2017    [ 0.7031; 154.6267]
## Viet Nam (National) 2017-2018          [ 26.4240; 75.0563]
## Ethiopia (Hetosa District) 2013         [ 75.5851; 191.0057]
## India (Tiruvallur District) 2015-2018   [ 14.5518; 55.7957]
## Ethiopia (Multiple Areas) 2012-2013     [ 44.2716; 262.3158]
##
## Number of studies: k = 53
## Number of observations: o = 1259916
## Number of events: e = 1269
##
##                               events          95%-CI
## Common effect model 100.7210 [95.3316; 106.4148]
## Random effects model 116.0037 [85.4497; 157.4657]
##

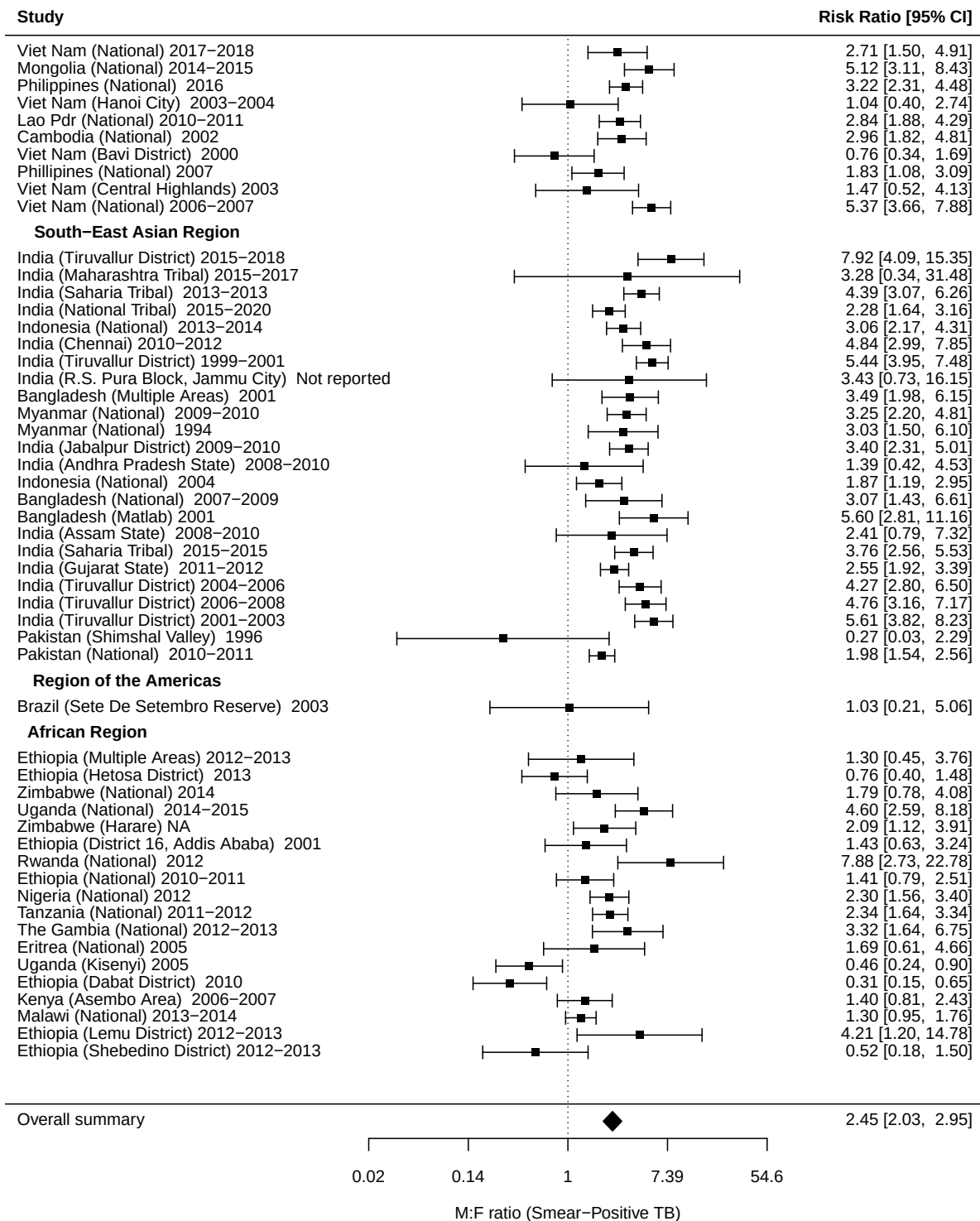
```

```
## Quantifying heterogeneity (with 95%-CIs):
## tau^2 = 1.1954; tau = 1.0934; I^2 = 95.0% [94.1%; 95.8%]; H = 4.48 [4.12; 4.86]
##
## Test of heterogeneity:
##      Q d.f.  p-value
## Wald 1041.70  52 < 0.0001
## LRT   932.27  52 < 0.0001
##
## Details of meta-analysis methods:
## - Random intercept logistic regression model
## - Maximum-likelihood estimator for tau^2
## - Calculation of I^2 based on Q
## - Logit transformation
## - Clopper-Pearson confidence interval for individual studies
## - Events per 100000 observations
```

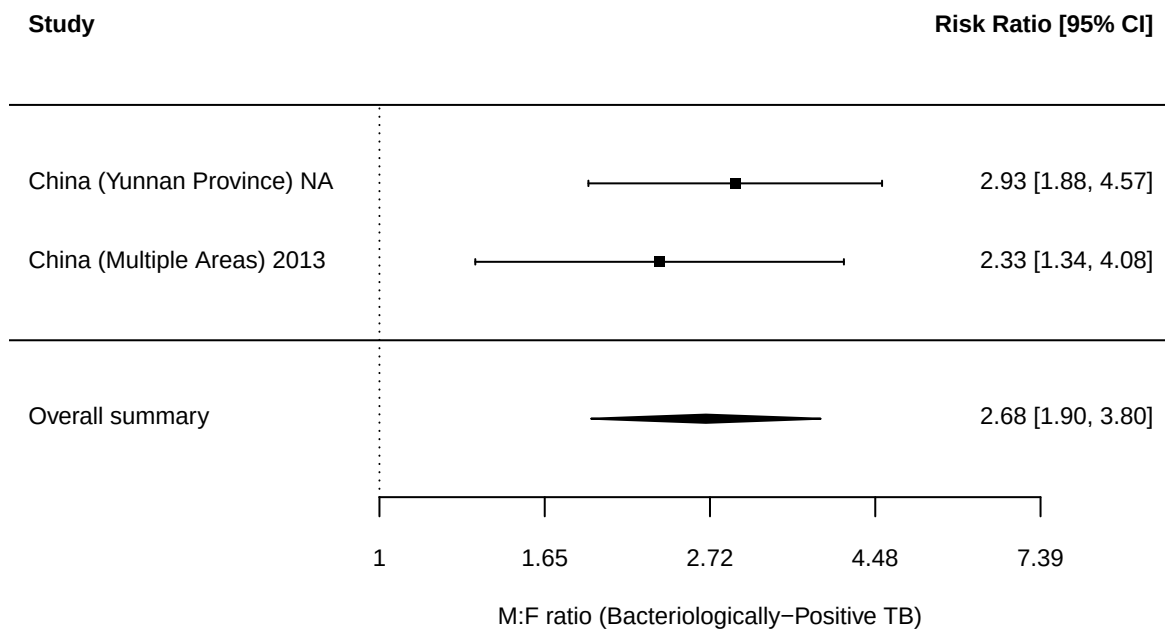
```
#####
##### MALE TO FEMALE RATIO OF SMEAR POSITIVE PREVALENCE #####
#####

### calculate order
prevSexSmear %<>% mutate(smear.order = n.smear.positive.tb.male/n.participants.male/
                           n.smear.positive.tb.female/n.participants.female)
smearPrevMF <- rma(ai=n.smear.positive.tb.male,
                  bi=n.participants.male - n.smear.positive.tb.male,
                  ci=n.smear.positive.tb.female,
                  di=n.participants.female - n.smear.positive.tb.female,
                  data=prevSexSmear, measure="RR",
                  slab=paste(figure.id.yr), method="REML")

par(cex=1, font=2)
metafor::forest(smearPrevMF,
                order = prevSexSmear$WHO.region,
                atransf=exp, cex=.85, xlab="M:F ratio (Smear-Positive TB)",
                mlab="Overall summary",
                psize=1, col="black", border="black",
                rows=c(3:20, 23, 26:49, 52:61))
par(cex=0.85, font=2)
text(-8.65, 50.5, "South-East Asian Region")
text(-8.85, 24.5, "Region of the Americas")
text(-9.65, 21.5, "African Region")
```

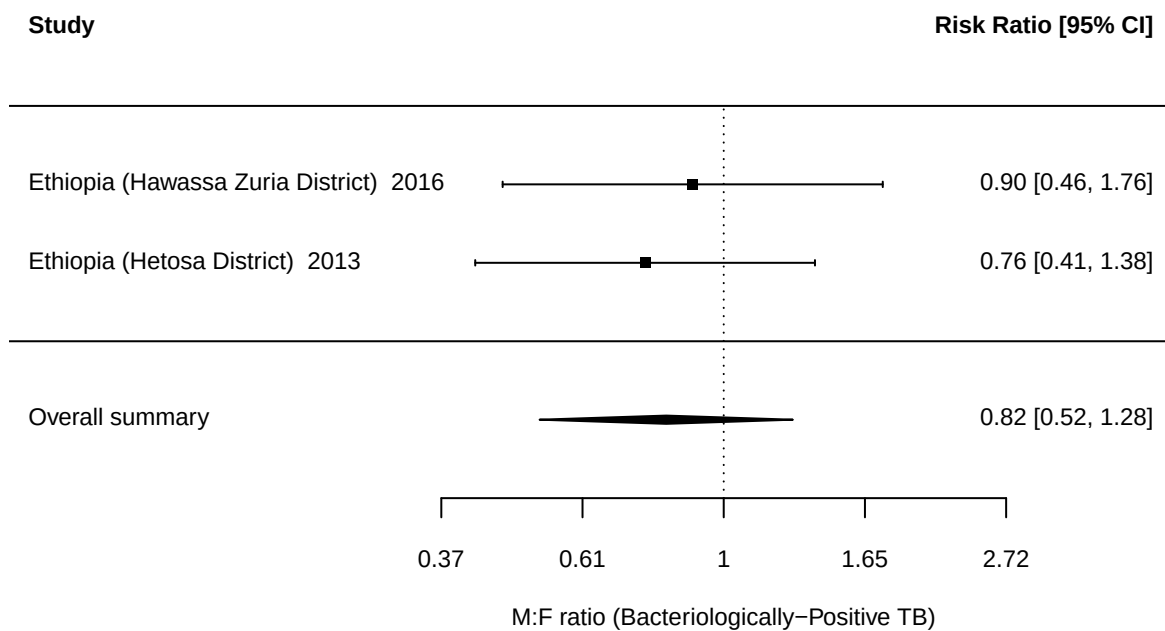


#COUNTRIES WITH MORE THAN ONE PREVALENCE SURVEY



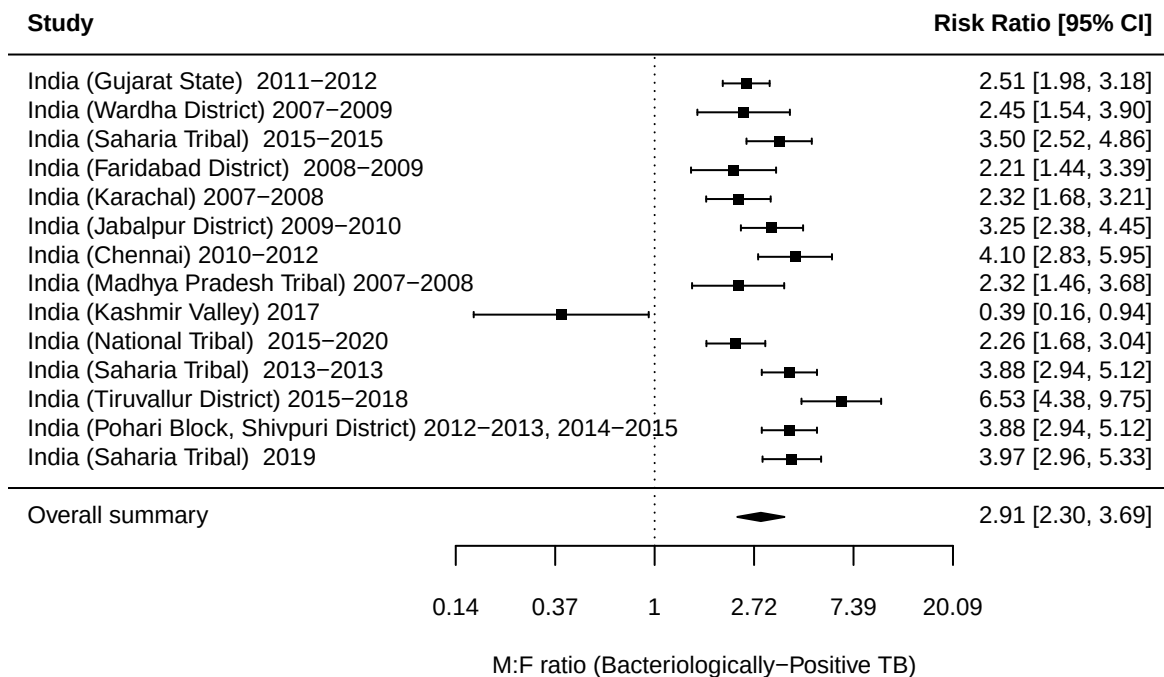
```
## $xlim
## [1] -1.12  2.40
##
## $alim
## [1] 0 2
##
## $at
## [1] 0.0 0.5 1.0 1.5 2.0
##
## $ylim
## [1] -2 5
##
## $rows
## [1] 1 2
##
## $cex
## [1] 0.75
##
## $cex.lab
## [1] 0.75
##
## $cex.axis
## [1] 0.75
##
## $ilab.xpos
## NULL
```

```
##
## $ilab.pos
## NULL
##
## $textpos
## [1] -1.12  2.40
##
## $areas
## [1] 0.40 0.35 0.25
```



```
## $xlim
## [1] -2.53  1.59
##
## $alim
## [1] -1  1
##
## $at
## [1] -1.0 -0.5  0.0  0.5  1.0
##
## $ylim
## [1] -2  5
##
## $rows
## [1] 1 2
##
```

```
## $cex
## [1] 0.75
##
## $cex.lab
## [1] 0.75
##
## $cex.axis
## [1] 0.75
##
## $ilab.xpos
## NULL
##
## $ilab.pos
## NULL
##
## $textpos
## [1] -2.53  1.59
##
## $areas
## [1] 0.40 0.35 0.25
```

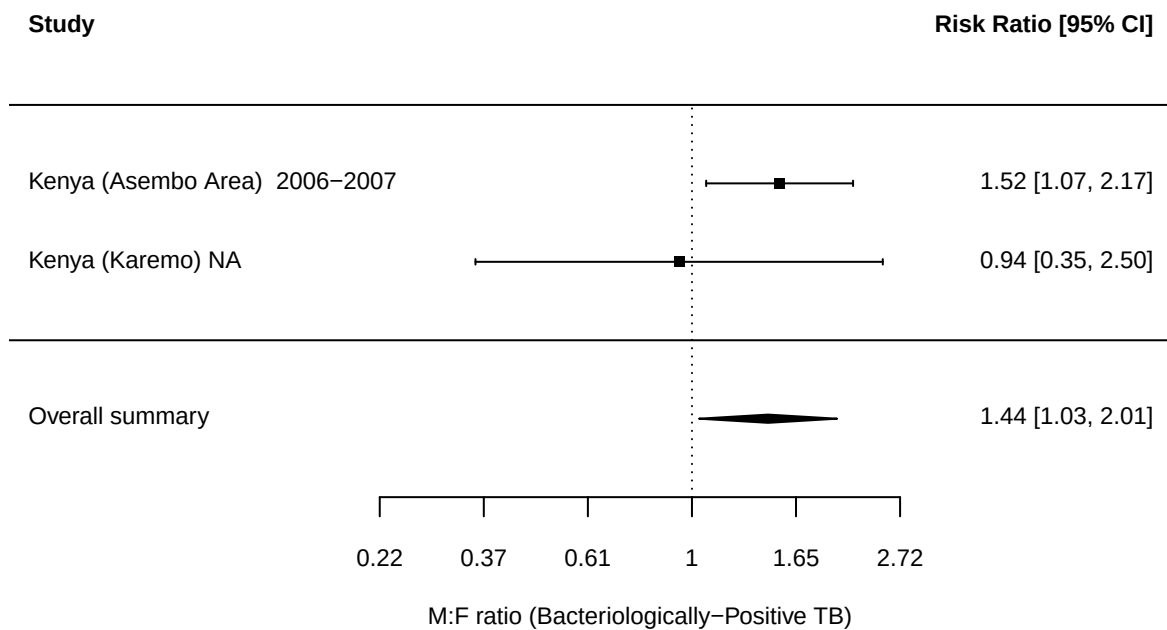


```
## $xlim
## [1] -6.5  5.2
##
## $alim
```

```

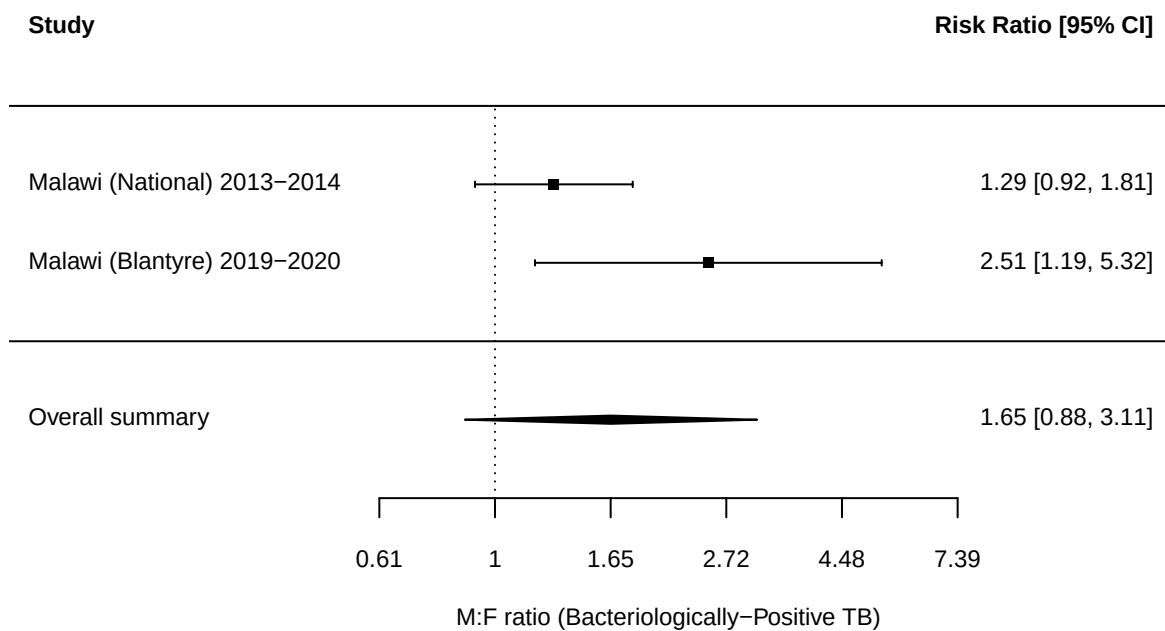
## [1] -2  3
##
## $at
## [1] -2 -1  0  1  2  3
##
## $ylim
## [1] -2 17
##
## $rows
## [1]  1  2  3  4  5  6  7  8  9 10 11 12 13 14
##
## $cex
## [1] 0.75
##
## $cex.lab
## [1] 0.75
##
## $cex.axis
## [1] 0.75
##
## $ilab.xpos
## NULL
##
## $ilab.pos
## NULL
##
## $textpos
## [1] -6.5  5.2
##
## $areas
## [1] 0.40 0.35 0.25

```



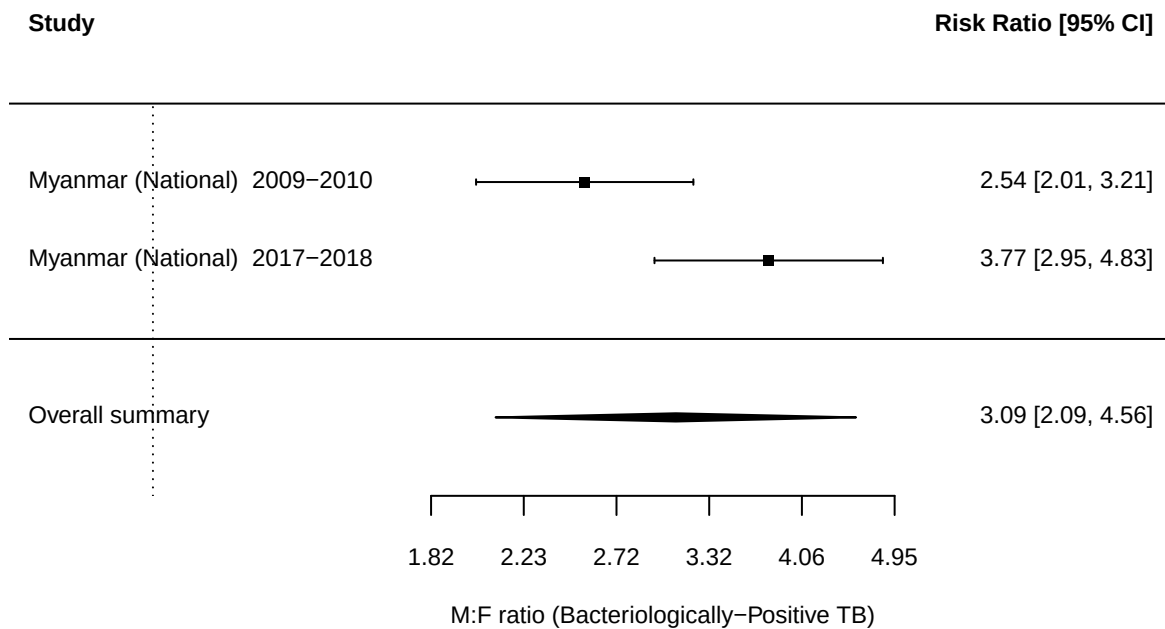
```
## $xlim
## [1] -3.28  2.31
##
## $alim
## [1] -1.5  1.0
##
## $at
## [1] -1.5 -1.0 -0.5  0.0  0.5  1.0
##
## $ylim
## [1] -2  5
##
## $rows
## [1] 1 2
##
## $cex
## [1] 0.75
##
## $cex.lab
## [1] 0.75
##
## $cex.axis
## [1] 0.75
##
## $ilab.xpos
## NULL
```

```
##
## $ilab.pos
## NULL
##
## $textpos
## [1] -3.28  2.31
##
## $areas
## [1] 0.40 0.35 0.25
```



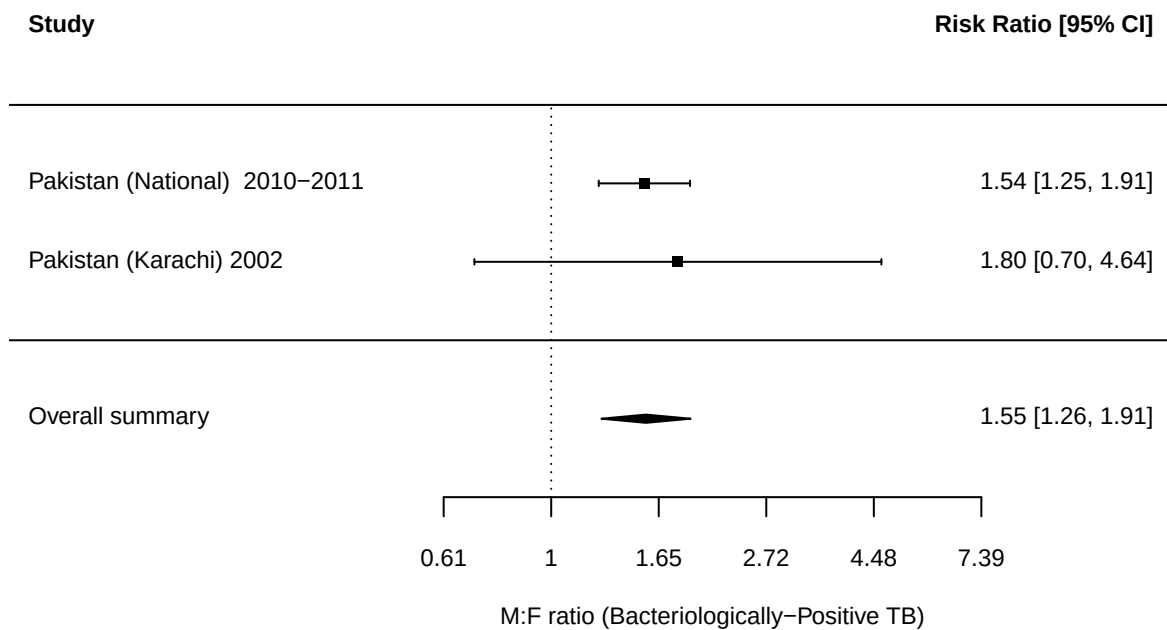
```
## $xlim
## [1] -2.10  2.93
##
## $alim
## [1] -0.5  2.0
##
## $at
## [1] -0.5  0.0  0.5  1.0  1.5  2.0
##
## $ylim
## [1] -2  5
##
## $rows
## [1] 1 2
##
```

```
## $cex
## [1] 0.75
##
## $cex.lab
## [1] 0.75
##
## $cex.axis
## [1] 0.75
##
## $ilab.xpos
## NULL
##
## $ilab.pos
## NULL
##
## $textpos
## [1] -2.10  2.93
##
## $areas
## [1] 0.40 0.35 0.25
```



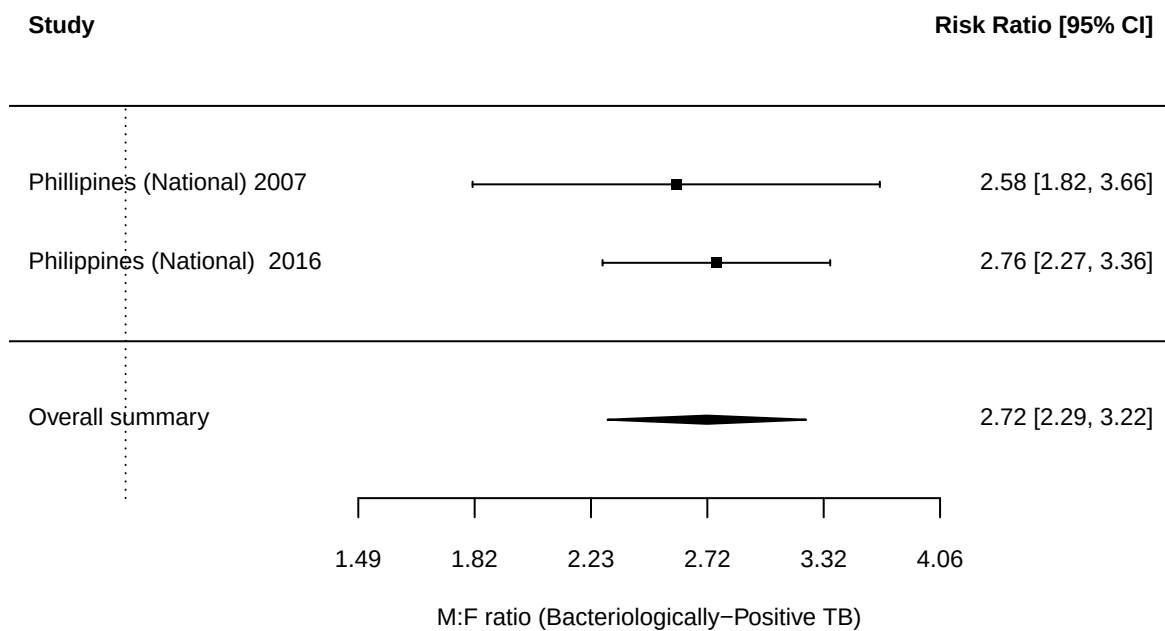
```
## $xlim
## [1] -0.31  2.20
##
## $alim
```

```
## [1] 0.6 1.6
##
## $at
## [1] 0.6 0.8 1.0 1.2 1.4 1.6
##
## $ylim
## [1] -2 5
##
## $rows
## [1] 1 2
##
## $cex
## [1] 0.75
##
## $cex.lab
## [1] 0.75
##
## $cex.axis
## [1] 0.75
##
## $ilab.xpos
## NULL
##
## $ilab.pos
## NULL
##
## $textpos
## [1] -0.31 2.20
##
## $areas
## [1] 0.40 0.35 0.25
```



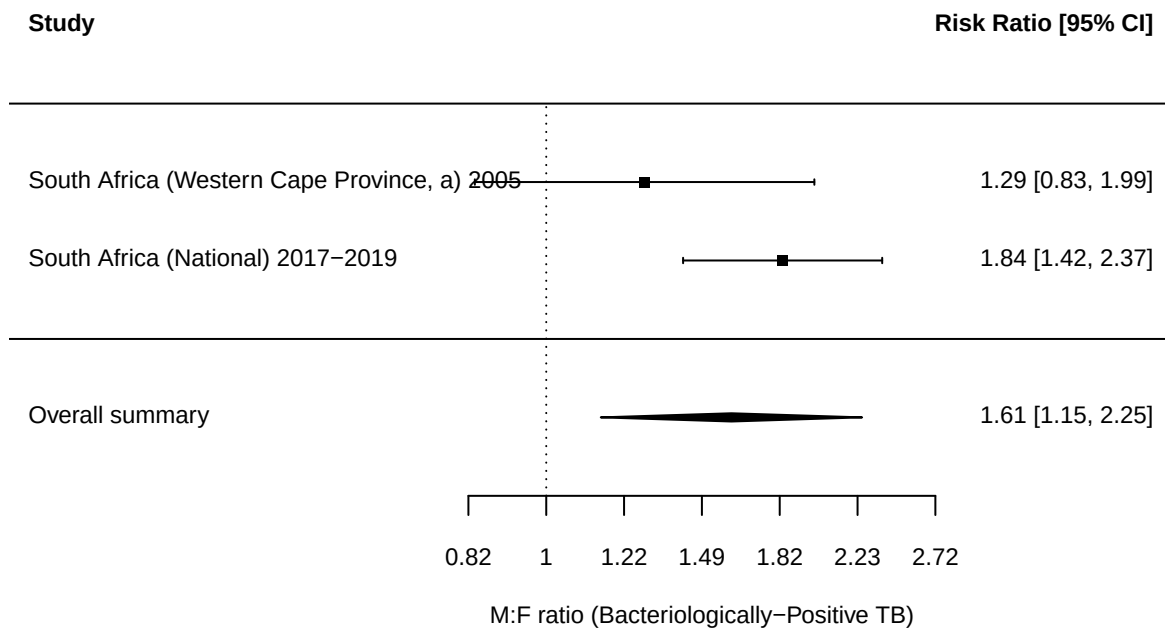
```
## $xlim
## [1] -2.52  2.89
##
## $alim
## [1] -0.5  2.0
##
## $at
## [1] -0.5  0.0  0.5  1.0  1.5  2.0
##
## $ylim
## [1] -2  5
##
## $rows
## [1] 1 2
##
## $cex
## [1] 0.75
##
## $cex.lab
## [1] 0.75
##
## $cex.axis
## [1] 0.75
##
## $ilab.xpos
## NULL
```

```
##
## $ilab.pos
## NULL
##
## $textpos
## [1] -2.52  2.89
##
## $areas
## [1] 0.40 0.35 0.25
```



```
## $xlim
## [1] -0.2 1.8
##
## $alim
## [1] 0.4 1.4
##
## $at
## [1] 0.4 0.6 0.8 1.0 1.2 1.4
##
## $ylim
## [1] -2 5
##
## $rows
## [1] 1 2
##
```

```
## $cex
## [1] 0.75
##
## $cex.lab
## [1] 0.75
##
## $cex.axis
## [1] 0.75
##
## $ilab.xpos
## NULL
##
## $ilab.pos
## NULL
##
## $textpos
## [1] -0.2 1.8
##
## $areas
## [1] 0.40 0.35 0.25
```

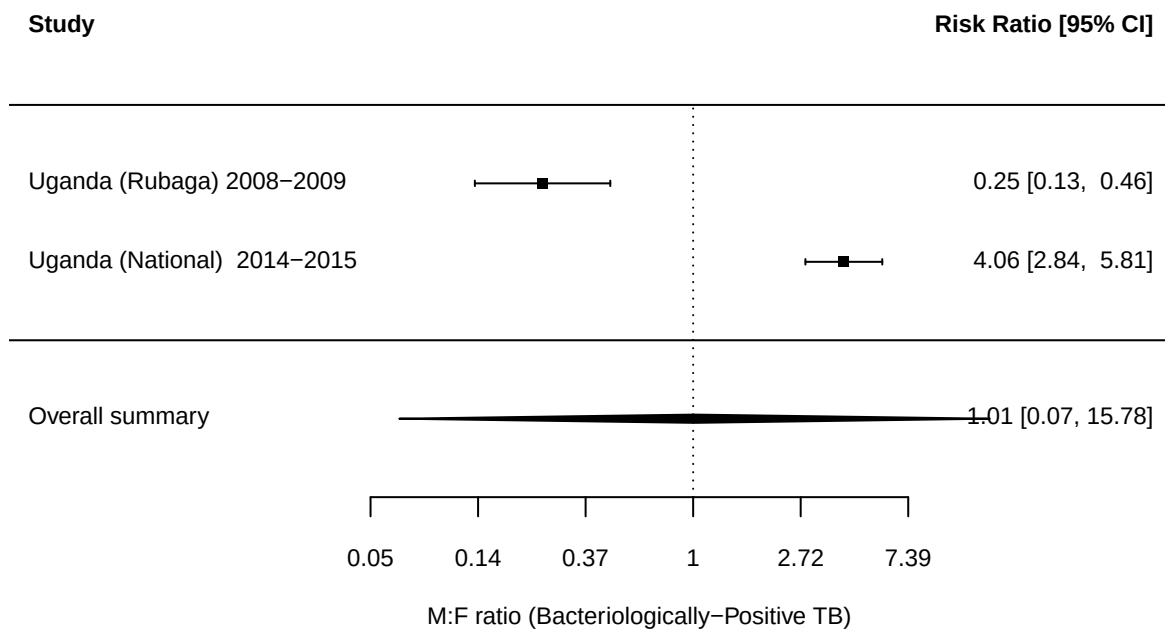


```
## $xlim
## [1] -1.38 1.61
##
## $alim
```

```

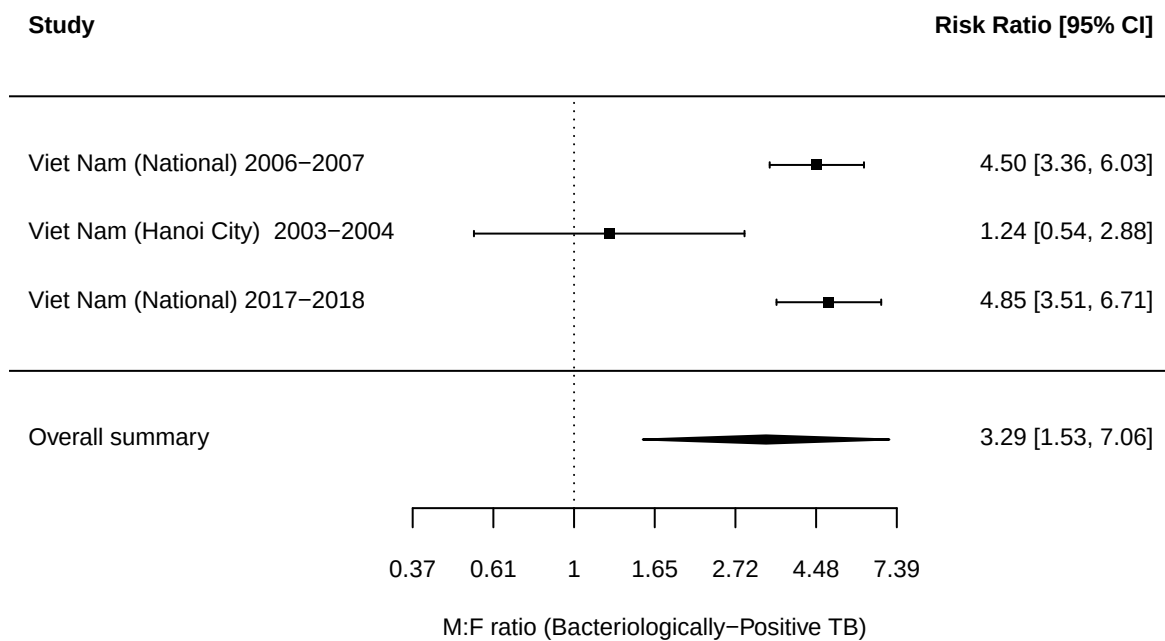
## [1] -0.2  1.0
##
## $at
## [1] -0.2  0.0  0.2  0.4  0.6  0.8  1.0
##
## $ylim
## [1] -2  5
##
## $rows
## [1] 1 2
##
## $cex
## [1] 0.75
##
## $cex.lab
## [1] 0.75
##
## $cex.axis
## [1] 0.75
##
## $ilab.xpos
## NULL
##
## $ilab.pos
## NULL
##
## $textpos
## [1] -1.38  1.61
##
## $areas
## [1] 0.40 0.35 0.25

```



```
## $xlim
## [1] -6.36  4.46
##
## $alim
## [1] -3  2
##
## $at
## [1] -3 -2 -1  0  1  2
##
## $ylim
## [1] -2  5
##
## $rows
## [1] 1 2
##
## $cex
## [1] 0.75
##
## $cex.lab
## [1] 0.75
##
## $cex.axis
## [1] 0.75
##
## $ilab.xpos
## NULL
```

```
##
## $ilab.pos
## NULL
##
## $textpos
## [1] -6.36  4.46
##
## $areas
## [1] 0.40 0.35 0.25
```



```
## $xlim
## [1] -3.50  3.71
##
## $alim
## [1] -1  2
##
## $at
## [1] -1.0 -0.5  0.0  0.5  1.0  1.5  2.0
##
## $ylim
## [1] -2  6
##
## $rows
## [1] 1 2 3
##
```

```
## $cex
## [1] 0.75
##
## $cex.lab
## [1] 0.75
##
## $cex.axis
## [1] 0.75
##
## $ilab.xpos
## NULL
##
## $ilab.pos
## NULL
##
## $textpos
## [1] -3.50 3.71
##
## $areas
## [1] 0.40 0.35 0.25
```

```
#####
##### UNIVARIATE META REGRESSION OF MF PREVALENCE RATIOS #####
#####
print("Univariate meta regression")
```

```
## [1] "Univariate meta regression"
```

```
### BY WHO REGION
# prevSexBact$WHO.region <- as.factor(prevSexBact$WHO.region)
#
# metab1 <- rma(ai=n.bacteriological.tb.male,
#               bi=n.participants.male - n.bacteriological.tb.male,
#               ci=n.bacteriological.tb.female,
#               di=n.participants.female - n.bacteriological.tb.female,
#               mods = regionWHO,
#               data=prevSexBact, measure="RR",
#               slab=paste(figure.id.yr), method="REML")
# metab1$b <- exp(metab1$b)
# metab1$ci.lb <- exp(metab1$ci.lb)
# metab1$ci.ub <- exp(metab1$ci.ub)
# summary(metab1)

### BY NATIONAL VS. SUBNATIONAL SURVEYS
prevSexBact %<>% mutate(national=ifelse(study.geography == "Nationally representative",
1, 0))

metab2 <- rma(ai=n.bacteriological.tb.male,
              bi=n.participants.male - n.bacteriological.tb.male,
              ci=n.bacteriological.tb.female,
              di=n.participants.female - n.bacteriological.tb.female,
              mods = ~national,
              data=prevSexBact, measure="RR",
```

```

slab=paste(figure.id.yr), method="REML")

print("Horton intrcpt=2.1930 , se=0.0893")

## [1] "Horton intrcpt=2.1930 , se=0.0893"

print(paste("Our intrcpt =", round(exp(metab2$b[1]),4), "se=", round(exp(metab2$se[1]),3)))

## [1] "Our intrcpt = 2.0653 se= 1.118"

print("Horton national=1.0142 ,se=0.1533")

## [1] "Horton national=1.0142 ,se=0.1533"

print(paste("Our national =", round(exp(metab2$b[2]),4), "se=", round(exp(metab2$se[2]),3)))

## [1] "Our national = 1.2396 se= 1.169"

summary(metab2)

##
## Mixed-Effects Model (k = 49; tau^2 estimator: REML)
##
##   logLik deviance      AIC      BIC     AICc
## -40.8633  81.7266  87.7266  93.2770  88.2847
##
## tau^2 (estimated amount of residual heterogeneity):    0.2496 (SE = 0.0610)
## tau (square root of estimated tau^2 value):           0.4996
## I^2 (residual heterogeneity / unaccounted variability): 89.76%
## H^2 (unaccounted variability / sampling variability):   9.77
## R^2 (amount of heterogeneity accounted for):           0.00%
##
## Test for Residual Heterogeneity:
## QE(df = 47) = 305.7858, p-val < .0001
##
## Test of Moderators (coefficient 2):
## QM(df = 1) = 1.8879, p-val = 0.1694
##
## Model Results:
##
##           estimate      se    zval    pval    ci.lb    ci.ub
## intrcpt      0.7253  0.1114  6.5122 <.0001   0.5070  0.9435 ***
## national     0.2148  0.1563  1.3740  0.1694 -0.0916  0.5212
##
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

```

### BY SURVEY START YEAR
prevSexBact %<>% mutate(start.year = study.start.year - 2001)
metab3 <- rma(ai=n.bacteriological.tb.male,
             bi=n.participants.male - n.bacteriological.tb.male,
             ci=n.bacteriological.tb.female,
             di=n.participants.female - n.bacteriological.tb.female,
             mods = ~start.year,
             data=prevSexBact, measure="RR",
             slab=paste(ffigure.id.yr), method="REML")

## Warning: 3 studies with NAs omitted from model fitting.

print("Horton intrcpt=5205946.3416, se=37.7545")

## [1] "Horton intrcpt=5205946.3416, se=37.7545"

print(paste("Our intrcpt =", round(exp(metab3$b[1]),4), "se=", round(exp(metab3$se[1]),3)))

## [1] "Our intrcpt = 1.6247 se= 1.225"

print("Horton start.year=0.9927, se=0.0188")

## [1] "Horton start.year=0.9927, se=0.0188"

print(paste("Our start.year =", round(exp(metab3$b[2]),4), "se=", round(exp(metab3$se[2]),3)))

## [1] "Our start.year = 1.0325 se= 1.018"

summary(metab3)

##
## Mixed-Effects Model (k = 46; tau^2 estimator: REML)
##
##   logLik  deviance      AIC      BIC      AICc
## -37.5899   75.1798   81.1798   86.5324   81.7798
##
## tau^2 (estimated amount of residual heterogeneity):      0.2308 (SE = 0.0585)
## tau (square root of estimated tau^2 value):             0.4804
## I^2 (residual heterogeneity / unaccounted variability): 89.19%
## H^2 (unaccounted variability / sampling variability):    9.25
## R^2 (amount of heterogeneity accounted for):             7.63%
##
## Test for Residual Heterogeneity:
## QE(df = 44) = 260.4296, p-val < .0001
##
## Test of Moderators (coefficient 2):
## QM(df = 1) = 3.3820, p-val = 0.0659
##
## Model Results:

```

```
##
##           estimate      se    zval    pval    ci.lb    ci.ub
## intrcpt      0.4853  0.2032  2.3888  0.0169   0.0871  0.8836  *
## start.year   0.0320  0.0174  1.8390  0.0659  -0.0021  0.0661  .
##
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

### BY LOW VS MODERATE/HIGH RISK OF BIAS

prevSexBact %<>% mutate(low.risk=ifelse(study.quality.summary ==
"Low risk of bias - Further research is very unlikely to change our confidence in the estimate",
1, 0))

metab7 <- rma(ai=n.bacteriological.tb.male,
             bi=n.participants.male - n.bacteriological.tb.male,
             ci=n.bacteriological.tb.female,
             di=n.participants.female - n.bacteriological.tb.female,
             mods = ~low.risk,
             data=prevSexBact, measure="RR",
             slab=paste(figure.id.yr), method="REML")

print("Horton intrcpt=2.4240 , se=0.1155")

## [1] "Horton intrcpt=2.4240 , se=0.1155"

print(paste("Our intrcpt =", round(exp(metab7$b[1]),4), "se=", round(exp(metab7$se[1]),3)))

## [1] "Our intrcpt = 2.1893 se= 1.123"

print("Horton low.risk=0.8481, se=0.1480")

## [1] "Horton low.risk=0.8481, se=0.1480"

print(paste("Our low.risk =", round(exp(metab7$b[2]),4), "se=", round(exp(metab7$se[2]),3)))

## [1] "Our low.risk = 1.0969 se= 1.171"

summary(metab7)

##
## Mixed-Effects Model (k = 49; tau^2 estimator: REML)
##
##    logLik deviance      AIC      BIC      AICc
## -41.6288  83.2576  89.2576  94.8080  89.8157
##
## tau^2 (estimated amount of residual heterogeneity):    0.2538 (SE = 0.0619)
## tau (square root of estimated tau^2 value):           0.5038
## I^2 (residual heterogeneity / unaccounted variability): 89.90%
## H^2 (unaccounted variability / sampling variability):  9.90
```

```
## R^2 (amount of heterogeneity accounted for):          0.00%
##
## Test for Residual Heterogeneity:
## QE(df = 47) = 306.2635, p-val < .0001
##
## Test of Moderators (coefficient 2):
## QM(df = 1) = 0.3425, p-val = 0.5584
##
## Model Results:
##
##           estimate      se    zval    pval    ci.lb    ci.ub
## intrcpt      0.7836  0.1164  6.7332 <.0001  0.5555  1.0116 ***
## low.risk      0.0925  0.1580  0.5852  0.5584 -0.2172  0.4022
##
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
### surveys with initial screening procedures requiring self-report of
### signs/symptoms vs.broader initial screening procedures.
```

```
prevSexBact %<>% mutate(symp.only = ifelse(screening.tests == "Symptom screen only",
                                           1, 0))
```

```
metab8 <- rma(ai=n.bacteriological.tb.male,
              bi=n.participants.male - n.bacteriological.tb.male,
              ci=n.bacteriological.tb.female,
              di=n.participants.female - n.bacteriological.tb.female,
              mods = ~symp.only,
              data=prevSexBact, measure="RR",
              slab=paste(figure.id.yr), method="REML")
```

```
print("Horton intrcpt=2.3384 , se=0.0829")
```

```
## [1] "Horton intrcpt=2.3384 , se=0.0829"
```

```
print(paste("Our intrcpt =", round(exp(metab8$b[1]),4), "se=", round(exp(metab8$se[1]),3)))
```

```
## [1] "Our intrcpt = 2.3834 se= 1.096"
```

```
print("Horton low.risk=0.8005, se=0.1622")
```

```
## [1] "Horton low.risk=0.8005, se=0.1622"
```

```
print(paste("Our low.risk =", round(exp(metab8$b[2]),4), "se=", round(exp(metab8$se[2]),3)))
```

```
## [1] "Our low.risk = 0.876 se= 1.196"
```

```
summary(metab8)
```

```
##
## Mixed-Effects Model (k = 49; tau^2 estimator: REML)
##
##   logLik  deviance      AIC      BIC      AICc
## -41.5244   83.0488   89.0488   94.5992   89.6069
##
## tau^2 (estimated amount of residual heterogeneity):      0.2521 (SE = 0.0615)
## tau (square root of estimated tau^2 value):              0.5021
## I^2 (residual heterogeneity / unaccounted variability): 89.87%
## H^2 (unaccounted variability / sampling variability):    9.88
## R^2 (amount of heterogeneity accounted for):              0.00%
##
## Test for Residual Heterogeneity:
## QE(df = 47) = 303.9734, p-val < .0001
##
## Test of Moderators (coefficient 2):
## QM(df = 1) = 0.5493, p-val = 0.4586
##
## Model Results:
##
##           estimate      se      zval      pval      ci.lb      ci.ub
## intrcpt      0.8685  0.0913   9.5096 <.0001   0.6895  1.0476 ***
## symp.only    -0.1324  0.1786  -0.7411  0.4586  -0.4825  0.2177
##
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
#### Surveys with diagnosis by smear microscopy vs.other diagnostic measures.
```

```
# prevSexBact %<>% mutate(smear.only = ifelse(screening.tests == "Symptom screen only",
#                                           1, 0))
#
# metab8 <- rma(ai=n.bacteriological.tb.male,
#               bi=n.participants.male - n.bacteriological.tb.male,
#               ci=n.bacteriological.tb.female,
#               di=n.participants.female - n.bacteriological.tb.female,
#               mods = ~symp.only,
#               data=prevSexBact, measure="RR",
#               slab=paste(figure.id.yr), method="REML")
#
# exp(metab8$b)
# exp(metab8$ci.lb)
# exp(metab8$ci.ub)
# summary(metab8)
```

```
##### |
### Close the PDF
# dev.off()
##### |
##### |
```